# 3D Anomaly Detection: A Survey

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***Abstract*—The rapid development of computer vision has brought 3D Anomaly Detection (3D-AD) wide attention in recent years. In this paper, we provide a comprehensive review of computer vision based 3D anomaly detection techniques from the perspectives of multimodal 3D-AD, point cloud 3D-AD, pose-agnostic 3D-AD, dataset, and benchmarks. In addition, we highlight several inherent or emerging challenges in image anomaly detection. The advantages and disadvantages of representative network architectures for different tasks are discussed. Finally, we summarize the research results and point out future research directions. For more resources, please visit https://github.com/M-3LAB/awesome-3d-anomaly-detection.**

***Index Terms*—3D anomaly detection, computer vision, point cloud, review, application.**

## I. Introduction

3D anomaly detection (AD) is a computer vision task that analyzes potential unexpected deformations in objects represented in three-dimensional spatial formats such as point clouds, voxels, depth maps, and meshes. It aims to detect and localize abnormal regions or defects in objects by identifying spatial structures, geometric features, or shape variations that deviate from normal patterns [1], [2], [3]. 3D anomaly detection has been widely studied as an effective approach for industrial quality inspection. Given that industrial quality inspection is a critical process to ensure products meet quality standards, its accuracy directly affects product quality and production efficiency [4], [5], [6].

In practice, traditional 2D image-based anomaly detection techniques have been extensively applied in industrial quality inspection scenarios, playing an important role in defect identification and quality control [7], [8], [9]. However, 2D image anomaly detection faces numerous challenges in practical applications, which can be summarized as follows:

1) Image acquisition and scanning are highly sensitive to illumination variations and occlusion.
2) Single-viewpoint inspection introduces inherent blind spots, which may result in missed detections.
3) It is difficult to accurately model explicit spatial geometric relationships using purely pixel-based representations.

Compared to the aforementioned limitations of traditional 2D image-based anomaly detection methods, 3D anomaly detection approaches can effectively alleviate these issues by explicitly modeling geometric structures and spatial relationships, thereby demonstrating significant advantages in industrial quality inspection applications. Meanwhile, as illustrated in Fig. 1, the 3D-AD research landscape can be systematically summarized along a pipeline of sensors, data modalities, methodological paradigms, and application scenarios: different 3D sensing devices give rise to RGB-D, point cloud, and multimodal representations, which are addressed by reconstruction-based or embedding-based paradigms and widely applied in automotive manufacturing, rail transit, medical inspection, integrated circuits, and advanced manufacturing.

Recently, 3D anomaly detection methods have reported strong performance on several benchmark datasets, highlighting the potential of 3D representations for accurate defect detection. Nevertheless, as an emerging research field, 3D anomaly detection still faces numerous challenges, particularly in achieving high precision, robustness, and deployment efficiency in real-world industrial environments. To the best of our knowledge, this work presents the first comprehensive survey of 3D anomaly detection, systematically summarizing the current research progress, technical methodologies, and application prospects of the field. The main contributions of this survey can be summarized below:

1) We comprehensively survey 3D anomaly detection methods under multiple modality settings, thoroughly examining detection model architectures across different data representations.

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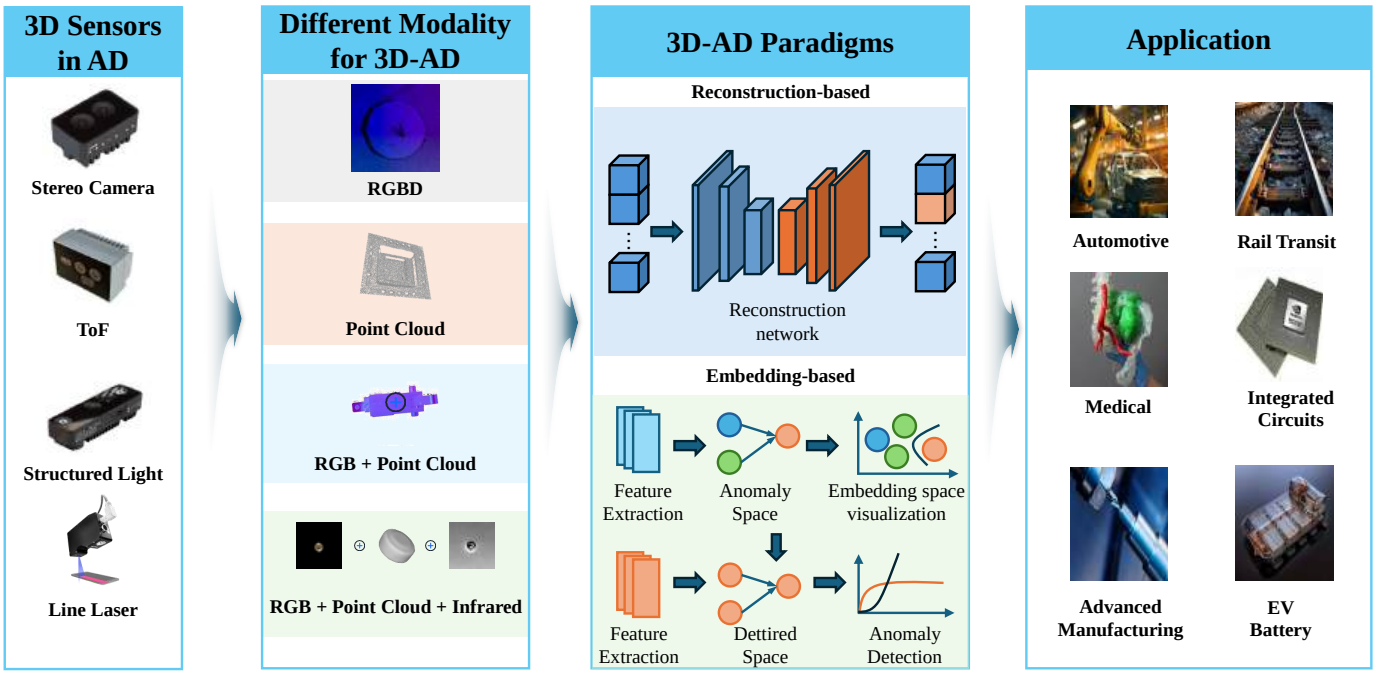



Fig. 1. **Overview of 3D-AD tasks:** Sensors → Data modalities → Paradigms → Applications, illustrating how 3D sensing leads to different representations and corresponding methodological choices in industrial anomaly detection.

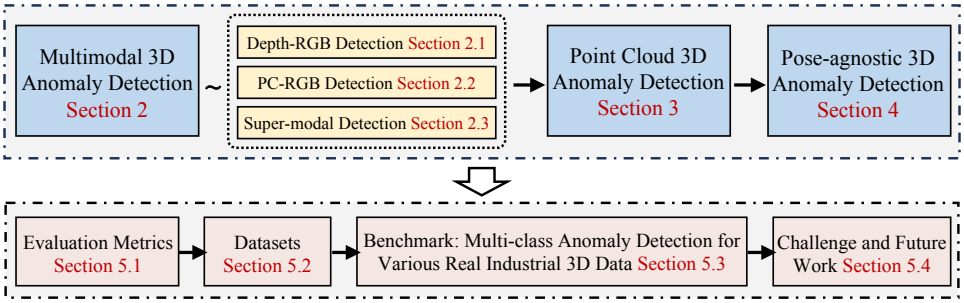



Fig. 2. Framework of this survey.

2) We systematically investigate 3D anomaly detection algorithms and datasets under diverse task configurations, with an emphasis on evaluating their practical viability in industrial environments.
3) We provide an in-depth analysis of key technical challenges in 3D anomaly detection and outline potential directions for future research.

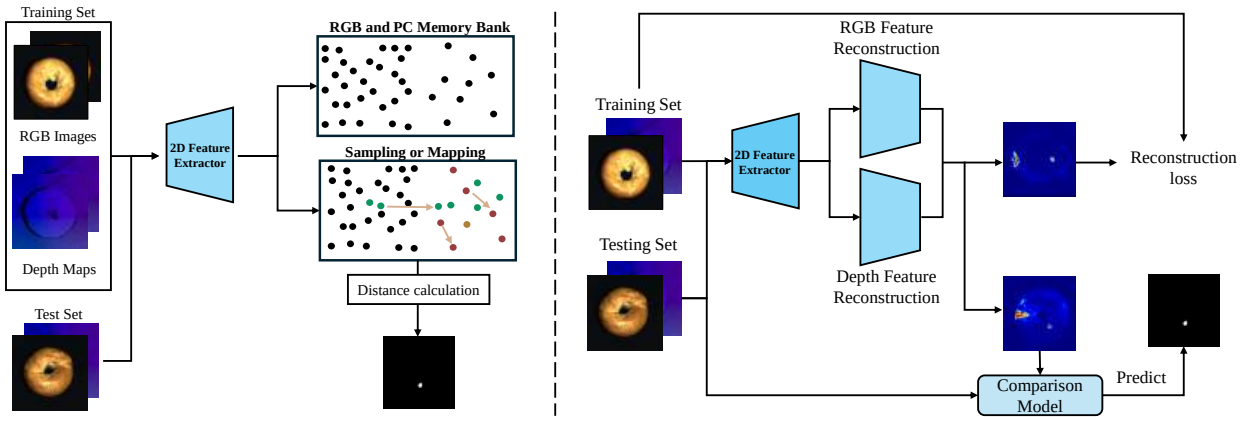
The rest of this paper is organized as shown in Fig. 2. In Sections 2, 3, and 4, we review 3D anomaly detection methods based on different modality settings. In Section 5, we comprehensively describe existing datasets and evaluation metrics. Subsequently, Section 6 categorizes methods according to different task settings and analyzes their applicability in real industrial scenarios. Finally, Section 7 discusses the remaining challenges and outlines future research directions for 3D anomaly detection.

## II. Multimodal 3D Anomaly Detection

Multimodal 3D anomaly detection (M3D-AD) is an anomaly detection method that uses three-dimensional spatial data as the core representation, integrates supplementary information from other modalities such as RGB images, textual descriptions, and sensor data, and improves accuracy through cross-modal feature learning. This field was initially proposed to extend the successful experience of image anomaly detection to 3D scenarios. Early datasets represented by MVTec3D-AD established the fundamental paradigm of multimodal 3D anomaly detection by providing pose-aligned point clouds and RGB image pairs. With the emergence of datasets like MulSen-AD, the available modal information has become increasingly rich, driving the development of higher-precision 3D anomaly detection methods. Based on this, we categorize multimodal 3D anomaly detection (M3D-AD) into three main categories according to the modality types used: Depth-RGB, Point Cloud (PC)-RGB, and Super-modal Detection (fusing more than two modalities).

### A. Depth-RGB Detection

*1) Problem Statement:* To compensate for the incomplete spatial information in image-based anomaly detection, depth-




**(a) Feature Embedding Method** **(b) Feature Reconstruction Method**

Fig. 3. Pipeline of Depth-RGB Detection. This pipeline illustrates how depth information and RGB appearance cues are jointly processed to model both geometric structures and surface textures for accurate anomaly detection.

RGB detection has been proposed to leverage complementary information from both depth maps and RGB images. The available data in the training phase contains depth map samples paired with corresponding RGB images, i.e., $P_{train} = \{(d_1, r_1), (d_2, r_2), \cdots, (d_n, r_n)\}$, where each $d_i$ denotes a normal depth map sample from the target category that represents 3D information through depth values, $r_i$ is the pose-aligned RGB image corresponding to $d_i$, and $n$ is the number of training samples. In the testing phase, the data to be detected includes both normal and anomalous depth-image pairs from the same category, i.e., $P_{test} = \{(\hat{d}_1, \hat{r}_1), (\hat{d}_2, \hat{r}_2), \cdots, (\hat{d}_m, \hat{r}_m)\}$, where each pair $(\hat{d}_j, \hat{r}_j)$ can be either normal or anomalous with corresponding labels $t_j \in \mathcal{T} = \{0, 1\}$ (0 denotes a normal sample and 1 denotes an anomaly).

*2) Feature Embedding Method:* The emergence of depth-RGB detection methods builds upon the successful practices in 2D visual detection. While depth maps differ fundamentally from RGB images in their semantic content, they offer remarkable flexibility in data representation. Depth information can be readily converted into formats such as pixel arrays or patches, which are structurally identical to those used for RGB images. The pipeline of the feature embedding method is shown in Fig. 3 (a).

This format compatibility enables the direct adoption of mature 2D visual backbones, such as ViT and ResNet [10], for depth-RGB anomaly detection. Nevertheless, the intrinsic semantic discrepancy between RGB and depth modalities limits the effectiveness of large-scale pre-trained models when applied to depth data. To alleviate this issue, AST [11] introduces an asymmetric student-teacher framework [12], [13], in which a Real-NVP-based normalising flow teacher [14], [15] and a feed-forward student are deliberately designed to exhibit inconsistent responses to out-of-distribution anomalies, thereby improving discriminability while reducing the over-generalisation commonly observed in symmetric distillation schemes. Distillation-based methods have become an important direction for efficient multimodal anomaly detection. LPFSTNet [16] and cross-modal distillation under missing-modality inference [17] highlight lightweight and robust knowledge transfer, while DAK-Net [18] and DCRDF-Net [19] further improve foreground-aware distillation, pseudo-anomaly modeling, and gated RGB–depth fusion. MMRD [20], following the reverse-distillation paradigm [21], instead reconstructs fused multimodal representations from a frozen teacher to enhance anomaly-sensitive cross-modal learning. These studies collectively show that multimodal distillation is most effective when it explicitly balances complementary knowledge transfer against modality interference.

*3) Feature Reconstruction Method:* Despite the substantial semantic disparities between depth maps and RGB images, their structural consistency enables the adaptation of most feature reconstruction methods. Notably, techniques such as pseudo-anomaly generation can be directly applied with depth-RGB alignment. The pipeline of feature reconstruction method is shown in Fig. 3 (b).

EasyNet [22], DBRN [23], and DRAIN [24] represent a class of lightweight RGB-D anomaly detection methods that emphasize deployment efficiency and adaptive multimodal fusion. Their shared motivation is that depth cues are not always beneficial in industrial scenarios, where noisy depth measurements may instead impair detection. Accordingly, these methods perform selective fusion to exploit depth information only when it provides complementary gains. EasyNet [22] adopts an attention-based entropy-guided fusion strategy [25], DBRN [23] introduces dynamic modality weighting, and DRAIN [24] further combines differentiated reconstruction objectives, including SSIM [26] and L2 losses, with a lightweight entropy-based dual-attention fusion design [27]. Overall, these methods show that effective RGB-D detection requires not only lightweight architectures, but also modality-adaptive fusion to suppress depth-induced noise.

Another major direction is reconstruction-driven anomaly detection, represented by 3DSR [28], DRÆM [29], 3D-PNAS [30], DAS3D [31], and 3DRÆM [32]. These methods reduce reliance on pretrained models by coupling reconstruction with anomaly simulation. In particular, 3DSR [28]

employs a Depth-Aware Discrete Autoencoder and uses Perlin noise [33], affine transformations, and grouped convolutions [34] to improve multimodal representation learning under limited depth data.

Recent advances further explore architectural optimization, anomaly generation, and unified multimodal modeling. 3D-ADNAS [35] revisits multimodal fusion through two-level neural architecture search with continuous relaxation [36], and its analysis based on Dempster-Shafer evidence theory [37] supports the robustness of well-designed fusion. Beyond anomaly generation, recent studies have also explored more unified multimodal integration. AnomalyHybrid [38] improves anomaly diversity through domain-agnostic GAN-based generation, while BridgeNet [39] and HGCF [40] emphasize more effective cross-modal fusion. In particular, BridgeNet connects point clouds, depth images, and pre-trained 2D backbones, further strengthened by describable-texture-based anomaly generation [41], whereas HGCF performs hierarchical geometry–color fusion to better exploit the complementarity between point-cloud and RGB information. An incremental unified multimodal anomaly detection framework [42] was developed from an information bottleneck [43] and MAMBA [44] perspective to improve denoising and continual adaptation and HFMM-Net adopted a Mamba-based hybrid fusion architecture for efficient multimodal industrial defect detection.. TRD [45] extends reverse distillation [21] to multimodal anomaly detection through multi-branch learning and explicit cross-modal interaction, alleviating the weakening of modality-specific anomalies during fusion.

In summary, RGB-D anomaly detection has evolved from straightforward multimodal aggregation toward selective fusion, reconstruction-guided learning, architecture-aware optimization, and cross-modal distillation. The central challenge is no longer simply how to combine RGB and depth, but how to exploit their complementarity without amplifying modality-specific noise or suppressing asymmetric anomalies [46].

### B. PC-RGB Detection

*1) Problem Statement:* To address the occlusion problem inherent in depth maps, PC-RGB detection has been proposed to leverage complementary information from both point clouds and RGB images. The available data in the training phase contains point cloud samples paired with corresponding RGB images, i.e., $P_{train} = \{(p_1, r_1), (p_2, r_2), \cdots, (p_n, r_n)\}$, where each $p_i$ denotes a normal point cloud sample from the target category, $r_i$ is the pose-aligned RGB image corresponding to $p_i$, and $n$ is the number of training samples. In the testing phase, the data to be detected includes both normal and anomalous point cloud-image pairs from the same category, i.e., $P_{test} = \{(\hat{p}_1, \hat{r}_1), (\hat{p}_2, \hat{r}_2), \cdots, (\hat{p}_m, \hat{r}_m)\}$, where each pair $(\hat{p}_j, \hat{r}_j)$ can be either normal or anomalous with corresponding labels $t_j \in \mathcal{T} = 0, 1$ (0 denotes a normal sample and 1 denotes an anomaly).

*2) Feature Embedding Method:* Point-level feature alignment is a central challenge in PC–RGB multimodal anomaly detection, especially for feature-embedding-based frameworks that require effective cross-modal fusion, as illustrated in Fig. 4(a). Existing methods in this line mainly follow either memory-bank or teacher–student paradigms. Early memory-bank approaches, such as BTF [47], adopt a relatively straightforward design by representing point clouds with handcrafted FPFH descriptors [48] and extracting RGB features with ResNet [10]. Separate memory banks are then constructed for the two modalities, and anomalies are identified through feature matching and dual-decision inference. While simple and effective, such methods lack explicit cross-modal interaction and therefore cannot fully exploit the complementarity between point cloud geometry and image appearance.

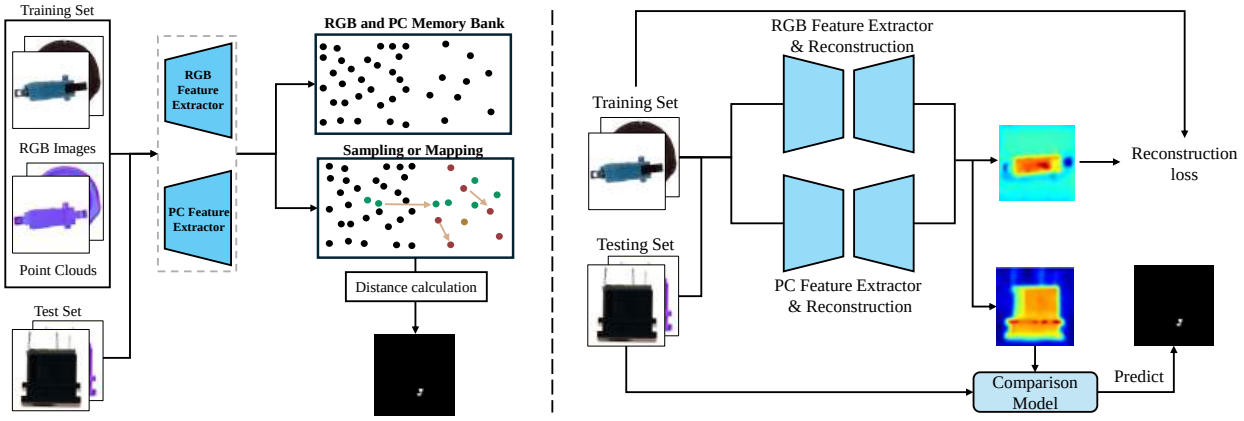
Subsequent studies have increasingly emphasized cross-modal fusion within the feature embedding space. M3DM [49] introduces a contrastive-learning-based fusion module [50] to jointly integrate point-cloud and RGB features and construct a fused feature distribution for improved anomaly discrimination. M3DM-NR [51] further enhances this framework by filtering noisy training samples through reference-image design and extending the representation to multi-scale features, thereby improving robustness in industrial settings. Along a related direction, the Shape-Guided method [52] builds a shape-feature memory bank based on signed distance representations and retrieves corresponding image features via K-nearest-neighbour search [53], enabling anomaly assessment through interpolated shape–image reference pairs and an attention-enhanced dual memory bank and memory-guided cross-modal scoring strategy were also introduced for multimodal industrial anomaly detection [54] . LSFA [55], by contrast, focuses on the domain-shift issue of pretrained multimodal models and introduces self-supervised feature adaptation with lightweight adapters to obtain more reliable task-specific representations.

More recent methods further strengthen multimodal representation learning through structured modeling. VLDFNet jointly models point-cloud and image information by constructing relational graphs across multiple views, disentangling shared and modality-specific semantics before fusion, and stabilizing the representation with vector quantization [56], [57], [58]. Memory-based multimodal anomaly detection has also been extended to few-shot and geometry-aware settings. CIF [59] leverages hypergraph structural priors [60], [61] to improve feature matching under extremely limited samples, whereas GPAD [62] incorporates point-cloud-derived geometric priors to strengthen detection of geometric defects.

Overall, PC–RGB feature embedding methods have evolved from independent unimodal memory construction toward more structured cross-modal representation learning. The key trend is to move beyond simple feature storage and retrieval, and instead explicitly model correspondence, complementarity, and structural consistency between geometric and visual modalities [63], [64].

*3) Feature Reconstruction Method:* Feature reconstruction methods in PC–RGB anomaly detection primarily aim to fuse 3D and 2D representations, typically employing generative modeling strategies such as transformer-based, flow-based, or GAN-based frameworks to enable token-level cross-modal interaction. The pipeline of feature reconstruction method is shown in Fig. 4 (b).

Inspired by the success of ViT [65] in 2D vision and




**(a) Feature Embedding Method** **(b) Feature Reconstruction Method**

Fig. 4. Pipeline of PC-RGB Detection. This pipeline illustrates how point cloud geometry and RGB appearance information are jointly exploited to model both spatial structures and surface characteristics for anomaly detection.

PointMAE [66] in 3D point cloud representation, recent PC–RGB anomaly detection methods have increasingly explored unified latent-space learning. CFM [67] maps point cloud and image features into a shared transformer space and performs mutual reconstruction across modalities, where reconstruction discrepancies serve as anomaly cues. FRM [68] and UCF [69] further strengthen this paradigm by improving cross-modal fusion and alignment, respectively through cross-attention and locality-aware feature integration.

A related direction seeks to unify cross-modal prediction and modality-specific reconstruction within a shared representation space. CPIR [70] introduces a latent intermediate space that supports both cross-modal prediction and intra-modal reconstruction, enabling parameter-shared multi-task learning for more robust anomaly detection. Uni-3DAD [71], inspired by the generative capability of GANs [72], instead emphasizes faithful normal reconstruction through adversarial learning. Flow-based methods such as MADFlow [73] further model cross-modal discrepancies explicitly, combining difference-compensation fusion with frequency-space enhancement to improve multimodal anomaly discrimination.

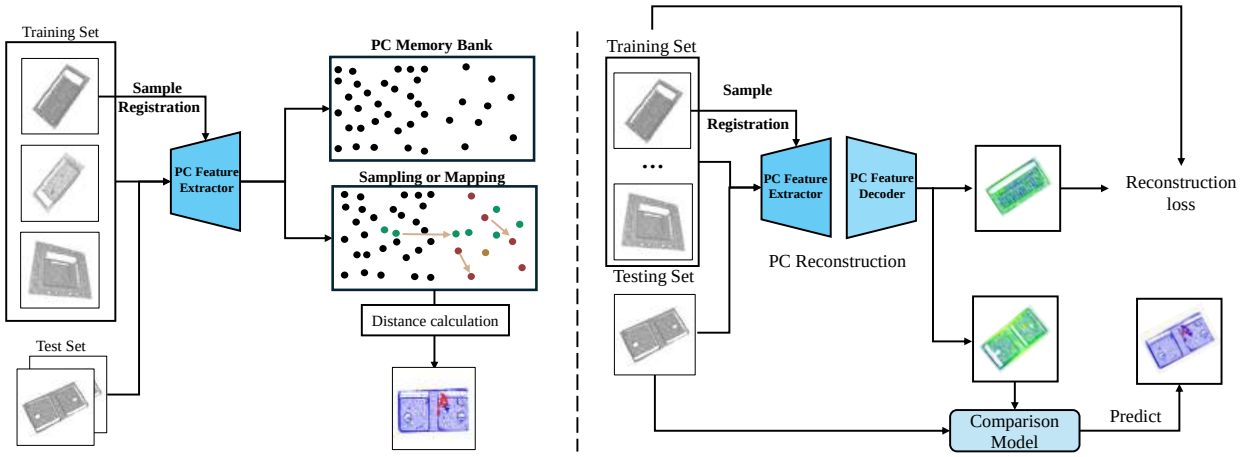
Recent efforts also place greater emphasis on efficiency and limited-data settings. MAFR [74] improves the efficiency–accuracy trade-off by learning a shared fusion space and reconstructing modality-specific features with decoupled attention-based decoders [75], thus avoiding the overhead of memory-bank-based methods. FIND [76], designed for few-shot anomaly detection, combines intra-modal reverse distillation with cross-modal feature mapping to improve both fine-grained localization and multimodal inconsistency detection under scarce normal samples. Cross-modal distillation was proposed to enable multimodal training and incomplete-modality inference for industrial anomaly detection [77].

Overall, this line of research extends PC–RGB anomaly detection from feature alignment toward unified latent-space modeling, where cross-modal fusion, reconstruction, and discrepancy learning are increasingly integrated into a common framework. A clear trend is that effective multimodal detection depends not only on aligning 2D and 3D features, but also on balancing reconstruction fidelity, cross-modal consistency, and data efficiency.

### C. Super-modal Detection

*1) Problem Statement:* To comprehensively capture object characteristics from multiple complementary perspectives, super-modal detection leverages information from more than two modalities simultaneously. Unlike depth-RGB or PC-RGB detection that relies on dual-modal fusion, super-modal detection integrates richer sensory information to achieve more robust anomaly identification. The available data in the training phase contains samples from multiple aligned modalities, i.e., $P_{train} = \{(m_1^1, m_1^2, \cdots, m_1^k), (m_2^1, m_2^2, \cdots, m_2^k), \cdots, (m_n^1, m_n^2, \cdots, m_n^k)\}$, where $k \geq 3$ denotes the number of modalities, each $m_i^j$ represents a normal sample from the $j$-th modality of the target category (e.g., $m_i^1$ could be an RGB image, $m_i^2$ a point cloud, and $m_i^3$ a thermal image from lock-in infrared thermography), and all modalities are spatially and temporally aligned for the $i$-th instance. Here $n$ is the number of training samples. In the testing phase, the data to be detected includes both normal and anomalous multi-modal samples from the same category, i.e., $P_{test} = \{(\hat{m}_1^1, \hat{m}_1^2, \cdots, \hat{m}_1^k), (\hat{m}_2^1, \hat{m}_2^2, \cdots, \hat{m}_2^k), \cdots, (\hat{m}_l^1, \hat{m}_l^2, \cdots, \hat{m}_l^k)\}$, where each multi-modal tuple $(\hat{m}_j^1, \hat{m}_j^2, \cdots, \hat{m}_j^k)$ can be either normal or anomalous with corresponding labels $t_j \in \mathcal{T} = \{0, 1\}$ (0 denotes a normal sample and 1 denotes an anomaly). The diversity of modalities and their complementary nature enable super-modal detection to capture defects that may be imperceptible in single or dual-modal settings.

*2) Preliminary Exploration:* Super-modal detection, which integrates information from more than two modalities, remains a relatively unexplored area in 3D anomaly detection. As a pioneering work in this direction, MulSen-TripleAD [78] proposes a multi-sensor anomaly detection baseline that combines RGB images, infrared images, and point clouds. Inspired by PatchCore [79] and M3DM [49], MulSen-TripleAD employs pretrained feature extractors to generate distinct representa-




**(a) Feature Embedding Method** **(b) Feature Reconstruction Method**

Fig. 5. Pipeline of Point Cloud 3D Anomaly Detection. This pipeline illustrates a typical workflow that operates directly on point cloud data to model 3D geometric structures and identify anomalous regions based on spatial deviations.

tions for each modality, specifically utilizing DINO [80] for RGB and infrared images and PointMAE [66] for point clouds. The method constructs separate memory banks $\mathcal{M}_{rgb}$, $\mathcal{M}_{ir}$, and $\mathcal{M}_{pt}$ for each sensor, storing normal feature representations during training following the PatchCore approach. During inference, each sensor's memory bank is used to compute an anomaly score based on the deviation of the test sample from the normal data. After obtaining the anomaly scores from each sensor, MulSen-TripleAD integrates these scores using a decision gating unit $\mathcal{G}_a$ inspired by the learnable One-Class Support Vector Machine (OCSVM) from M3DM [49]. The final object-level anomaly score is calculated by aggregating individual sensor scores through this gating mechanism, enabling effective multi-sensor fusion for robust anomaly detection across diverse industrial scenarios.

## III. Point Cloud 3D Anomaly Detection

Point cloud 3D anomaly detection (PC3D-AD) performs anomaly detection purely based on point cloud data, utilizing techniques such as point cloud registration, feature representation, and reconstruction to identify anomalous points or structures. This approach can fully exploit the 3D geometric characteristics of point clouds while avoiding the complexity of multimodal fusion. Current mainstream PC3D-AD methods are primarily divided into three categories: feature embedding-based methods, which leverage point cloud feature representations for anomaly identification; reconstruction-based methods, which detect anomalous patterns through point cloud completion or reconstruction; and zero-shot methods, which have recently demonstrated greater industrial potential through cross-category transfer capabilities.

### A. Problem Statement

In the training phase of PC3D-AD, the available data contains only normal point cloud samples, denoted as $P_{train} = \{p_1, p_2, \cdots, p_n\}$, where each $p_i$ represents a normal point cloud sample from the target category and $n$ is the number of training samples. In the testing phase, the data includes both normal and anomalous point cloud samples from the same category, represented as $P_{test} = \{\hat{p}_1, \hat{p}_2, \cdots, \hat{p}_m\}$, where each $\hat{p}_j$ can be either normal or anomalous with corresponding ground-truth labels $t_j \in \mathcal{T} = \{0, 1\}$ (0 denotes a normal sample and 1 denotes an anomaly).

### B. Feature Embedding Methods

As shown in Fig. 5 (a), feature embedding methods construct semantic normal feature distributions using pre-trained feature extractors or mathematical operators to represent point cloud features. Anomaly detection is then achieved by comparing the differences between test point clouds and these normal feature distributions. Based on the feature extraction approaches, we categorize these methods into three types: handcrafted mathematical descriptors, deep feature extractors, and multi-view feature representations.

*1) Handcrafted Mathematical Descriptors:* Early research on 3D point cloud anomaly detection was largely constrained by the unordered nature of point clouds and the lack of effective pretraining protocols, and therefore relied heavily on handcrafted geometric descriptors for feasibility validation. As a pioneering attempt, 3D-ST [81] introduced a student–teacher framework with custom local feature aggregation, where local geometric structures are modeled via k-nearest-neighbor graphs [82] and the teacher is initialized by self-supervised pretraining based on RandLA-Net [83]. Building on this line, BTF [47] and PatchCore-3D [84] systematically demonstrated the effectiveness of feature embedding methods based on Fast Point Feature Histograms (FPFH) [48], highlighting the importance of rotation-invariant geometric representations for 3D anomaly detection.

To examine whether mature 2D anomaly detection paradigms [85], [86], [21] can be extended to 3D settings, CPMF [87] combines FPFH descriptors with complementary features from multi-view rendering [88], validating the benefit of multimodal feature fusion. From a more principled

perspective, Fence Theory [89] further argues that effective feature embedding requires alignment between feature distributions and semantic consistency. Addressing the limited geometric sensitivity of sparse-sampling-based methods, MiniShift [90] introduces multi-scale neighborhood descriptors and a lightweight aggregation framework, enabling real-time [91] anomaly detection on high-resolution point clouds.

Recent extensions further adapt handcrafted-feature-based pipelines to more specialized scenarios. 3DMulti-FPFHI [92] enhances FPFH-based embedding with intensity cues for crack and water-ingress detection in ageing structures, demonstrating the value of multimodal geometric descriptors under limited data. In parallel, Untrained Machine Learning [93] targets single-sample anomaly detection in personalized manufacturing through three training-free frameworks, namely Latent Variable Inference, Decomposition, and Local Geometry, and achieves competitive performance by exploiting geometric priors and sparse learning [94] with substantially improved inference efficiency.

Overall, these early studies established the feasibility of 3D anomaly detection through handcrafted and training-light representations, and showed that robust geometric description, semantic alignment, and efficiency-aware design are foundational to later learned 3D anomaly detection frameworks.

*2) Deep Feature Extractors:* While handcrafted descriptors offer interpretability, they are inherently limited in capturing complex geometric patterns, which motivates the use of deep feature extractors for more discriminative 3D anomaly representation. Early deep feature embedding methods were mainly developed within teacher–student and memory-bank paradigms. TS-AD [95] adopts a pre-trained PointNet teacher to guide a student network toward category-specific normal representations through feature mimicry. Inspired by PatchCore [79], Reg3D-AD [96] further introduces memory-bank-based feature embedding for point clouds by combining PointMAE [66] features from RANSAC-registered point clouds [97] with point-level geometric information, while PointCore [98] improves this framework through a simplified memory-bank design and normalized ranking for faster inference.

A subsequent line of research focuses on improving representation discriminability and robustness. Group3AD [99] enhances feature separability through contrastive-learning-based group clustering, whereas ISMP [100] transforms point cloud structures into 2D plane representations and leverages EfficientNet [101] to complement global geometric encoding. To support multi-class anomaly detection within a unified embedding space, GLFM [102] combines anomaly synthesis and global–local feature matching [82] to construct category-shared memory representations. Reg2Inv [103] further incorporates point cloud registration into feature learning, jointly modeling geometric correspondence and anomaly-sensitive representation with RIConv++ [104], KPConv-FPN [105], and Geometric Transformer [106]. Along a similar direction, RIF [107] improves robustness by explicitly mapping point clouds into a rotation-invariant feature space. Template3D-AD [108] and ITNM [109] used center-point-based template matching or incremental template neighborhood matching to detect anomalies in point clouds.

Beyond standard anomaly detection, feature embedding methods have also been extended to more challenging settings. Deep Subspace Learning [110] addresses intra-class variation, inter-class similarity, and unseen anomaly types through subspace-based modeling built on a lightweight PointNet encoder [111]. RoDA [112], by contrast, targets performance degradation under distribution shift by introducing optimal-transport-based distribution alignment [21], [113] and robust Sinkhorn optimization [114] on top of memory-bank-based detection, thereby improving adaptation to target-domain data without modifying source training.

Taken together, deep feature embedding methods have moved 3D point cloud anomaly detection beyond simple teacher–student imitation toward memory-based, contrastive, registration-aware, and distribution-adaptive representation learning. The broader trend is that effective 3D anomaly detection depends not only on stronger feature extractors, but also on embedding spaces that are discriminative, transformation-consistent, and robust to domain variation.

*3) Multi-view Feature Representations:* Multi-view representation provides an effective bridge between mature 2D image anomaly detection methods and 3D point cloud scenarios by exploiting complementary observations from different perspectives. Early studies mainly validated the benefit of combining geometric and view-based cues. For instance, CPMF [87] integrates FPFH descriptors with multi-view rendering features to demonstrate the advantage of multimodal feature fusion, while ISMP [100] adopts an internal multi-view strategy that projects point cloud structures into multiple 2D plane representations and uses EfficientNet for feature extraction to enhance global representation.

Subsequent methods place greater emphasis on fine-grained geometric alignment between point clouds and rendered views. Looking3D [115] departs from conventional global multi-view fusion by learning precise correspondences between query images and multi-view representations through a correspondence matching transformer [116], where sparse cross-attention and 3D positional encoding [65] enable geometry-aware anomaly detection. By contrast, MVR [117] takes a more radical perspective by fully converting high-resolution point clouds into multi-view rendered images and directly leveraging mature 2D anomaly detection pipelines, without relying on native point cloud features. Relatedly, 2M3DF [118] combines multi-view RGB images and point clouds through multimodal fusion to further improve 3D industrial defect detection.

These methods show that multi-view representation serves as an effective bridge for transferring mature 2D anomaly detection techniques to 3D settings, whether by complementing point cloud features, enabling cross-view correspondence, or directly replacing native 3D representations [119], [120].

### C. Reconstruction-based Methods

In contrast to feature embedding methods that compare test samples against stored normal distributions, reconstruction-based methods learn to reconstruct normal patterns and detect anomalies through reconstruction errors. These methods utilize

pre-trained feature extractors to represent group-level or point-level features, then employ a proxy model to process and reconstruct these features, as shown in Fig. 5 (b). Given the reconstruction objective, the proxy model can differentially handle features corresponding to normal and anomalous structures, enabling anomaly detection through reconstruction discrepancy analysis. Based on different reconstruction strategies, we categorize these methods into encoder-decoder reconstruction, offset-based prediction, and shape descriptor representation.

*1) Encoder-Decoder Reconstruction:* To leverage the achievements and experience from 2D anomaly detection methods, a FoldingNet-based [121] autoencoder architecture was first introduced [122] for 3D scenarios. This approach employs a generative modeling paradigm to produce anomaly-free point clouds corresponding to test samples, detecting anomalies through comparison between input and reconstructed outputs.

Reconstruction-driven methods further improve 3D point cloud anomaly detection by explicitly restoring anomaly-free geometry. IMRNet [123] introduces group-wise iterative masked reconstruction [124] for more precise anomaly removal, while DUS-Net [125], DAUP [126], and IAENet [127] respectively enhance scalability, point-cloud homogeneity, and robustness. This line of work indicates that high-quality normal reconstruction requires jointly considering local precision, resolution efficiency, and structural consistency. However, these methods fall short in leveraging the capabilities of models trained on large-scale data. Drawing on the successful practices of pre-trained feature extractors, mapping raw point clouds to discriminative high-dimensional feature spaces through PointMAE [66] or PointBERT [128] can significantly improve detection performance. This approach demonstrates superior potential particularly in multi-class anomaly detection tasks [129]. MC3D-AD [130] first exploits this direction by employing geometry-aware attention mechanisms [131] to process pre-trained features for unified anomaly detection across multiple categories. Extending this approach, C3D-AD [132] employs Kernel Attention with Learnable Advisor to further adapt multi-class anomaly detection methods for continual learning scenarios, better meeting evolving industrial requirements. More recently, Anomaly-Aware Comparative Transformer (AACoT) [133] was proposed for robust point cloud anomaly detection. Vote3D-AD [134] combined varied defect synthesis with differentiable vote-clustering for unsupervised point-cloud anomaly localization. It utilizes a normal-to-anomaly comparative attention mechanism to highlight defective regions and reduce false positives, while a Siamese decoder refines localization through contrastive prediction. However, the method's reliance on high-quality normal references incurs significant memory overhead.

*2) Offset-based Prediction:* Offset-based methods model anomaly detection as a point displacement prediction task, offering an alternative perspective to traditional reconstruction approaches. Initially, R3D-AD [135] employs Denoising Diffusion Probabilistic Models (DDPMs) [136] to progressively remove potential anomalies in point clouds, achieving effective anomaly detection. However, diffusion-based methods impose high requirements for point cloud downsampling and memory usage, limiting their practical deployment. To address these limitations, PO3AD [137] employs the efficient MinkUNet34C [138], [139] to predict offsets for each point without any downsampling. This approach achieves both efficient inference speed and balanced memory requirements, yielding highly promising results for industrial applications. Recently, MC4AD [140] extends PO3AD by reconsidering the origins of anomalies from a mechanical perspective rather than solely detecting them through morphological approaches, achieving notable progress in understanding and localizing defects. Building upon PO3AD, several methods further enhance point-level descriptive capability. CASL [141] improves reconstruction quality by exploiting point-level curvature cues and selective masking, thereby encouraging the separation of normal and abnormal patterns in low-level geometric space, while further supporting the importance of curvature for 3D anomaly detection [142]. In addition, UCA-3DL [143] addresses the distribution shift between training and testing data via point-level test-time adaptation, leading to more precise point-wise representation and anomaly localization performance.

*3) Shape Descriptor Representation:* Beyond purely data-driven representations, another line of research seeks more interpretable and principled point-level anomaly description through explicit shape modeling, structural priors, and semantic representation learning. Inspired by Neural-Pull [144], SDF-based methods combine PointNet [111] with neural implicit functions [145] to characterize normal surface geometry and identify out-of-distribution anomalous points. Along a related direction, 3D-CSAD [146] addresses anomaly-scarce manufacturing scenarios through an untrained framework that decomposes complex surfaces into simple components and performs single-sample anomaly detection via robust principal component analysis [147]. PASDF [148] further extends this line by learning continuous pose-invariant shape representations, thereby unifying 3D anomaly detection and repair.

A parallel direction emphasizes richer semantic representation for point-level anomaly analysis. PLANE [149] transfers the generalization capability of pre-trained point-language models to 3D anomaly detection through multi-modal prompts, enabling strong cross-category performance with a single model. 3DKeyAD [150] instead focuses on high-resolution industrial inspection and improves localization through registration-based multi-prototype alignment and cluster-wise discrepancy analysis. 3D-SONAR [151] takes a training-free perspective and introduces a self-organizing framework for anomaly ranking in 3D point clouds. These studies suggest that point-level anomaly detection can be strengthened not only by data-driven feature learning, but also by more explicit geometric modeling, stronger semantic priors, and training-efficient design [152], [153].

### D. Zero-shot Methods

To meet the increasing industrial demand for cross-category transfer without retraining, a growing body of work has explored zero-shot 3D anomaly detection. An early attempt employs GPT-4V [154] to validate the potential of large

vision-language models in zero-shot 3D industrial anomaly detection [155], suggesting that large-scale multimodal priors can be transferred to unseen categories. Along this line, PointAD [156], built upon CLIP-style generalization for point clouds [157], and its extension PointAD+ [158] adapt vision-language priors to 3D anomaly recognition through point-pixel representation learning, prompt optimization, and geometry-aware spatial modeling. Relatedly, 3DzAL [159] focuses on non-overlapping train–test categories and improves cross-category detection through pseudo-anomaly generation and patch-level contrastive learning. More recently, GS-CLIP [160] further strengthens this direction by introducing geometry-aware prompts and synergistic view representation learning.

Another important strategy is to reformulate 3D anomaly detection as a 2D or vision-language modeling problem through multi-view projection. MVP [161] projects point clouds into multi-view depth images and directly leverages pre-trained vision-language models for anomaly detection. CLIP3D-AD [162] follows a similar rationale, combining adapter-based CLIP fine-tuning [163], multi-view projection, and anomaly synthesis to address data scarcity in few-shot 3D anomaly detection. Extending this idea to broader industrial scenarios, Customizable-VLM [164] unifies images, point clouds, and videos into a common 2D representation and incorporates expert-guided multimodal prompting, demonstrating the flexibility of vision-language models for anomaly detection and reasoning across modalities.

A parallel line of research emphasizes multimodal collaboration and sample-level relation modeling in zero-shot settings. MCL-AD [165] introduces multimodal collaborative learning for zero-shot 3D anomaly detection, while MuSc-V2 [166] improves zero-shot multimodal anomaly classification and segmentation through mutual scoring among unlabeled samples. Collectively, these methods indicate that zero-shot 3D anomaly detection is evolving from direct transfer of pretrained multimodal priors toward more structured paradigms that jointly exploit prompt learning, geometric modeling, multi-view projection, and multimodal collaboration.

## IV. Pose-agnostic 3D Anomaly Detection

Pose-agnostic 3D anomaly detection (PA3AD) is a detection method proposed to address issues in traditional point cloud anomaly detection methods, such as slow feature extraction and high sensitivity to viewpoint poses. This method leverages known or estimated pose information to query standard samples from corresponding viewpoints in 3D data. It identifies anomalies based on the differences between the query results and the samples to be detected.

### A. Problem Statement

For PA3AD, the available data in the training phase contains image samples that may optionally include pose information, i.e., $P_{train} = \{(r_1, \theta_1), (r_2, \theta_2), \cdots, (r_n, \theta_n)\}$, where each $r_i$ denotes a normal RGB image sample from the target category, $\theta_i$ represents the optional pose information (which may be available or unavailable), and $n$ is the number of training samples. The model reconstructs a 3D object representation (such as 3D Gaussian Splatting) from these training images. In the testing phase, given a query image $\hat{r}_j$, the model first estimates the pose $\hat{\theta}_j$, then renders the expected normal image under that pose, and finally determines anomalies by comparing the query image with the rendered normal image. Each test sample has a corresponding label $t_j \in \mathcal{T} = \{0, 1\}$ (0 denotes a normal sample and 1 denotes an anomaly). The structure of the Pose-agnostic 3D Anomaly Detection models is depicted in Fig. 6.

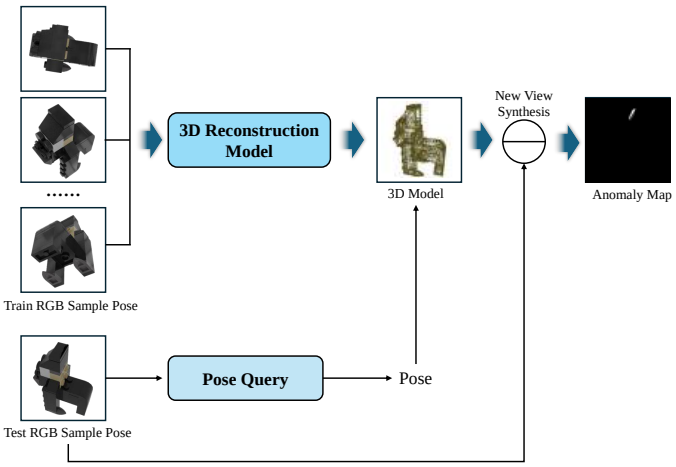



Fig. 6. Architecture of Pose-agnostic 3D Anomaly Detection models. This figure summarizes representative model designs that aim to achieve robust anomaly detection under pose variations, reducing sensitivity to object viewpoint, orientation, and spatial alignment.

### B. Preliminary Exploration

To remove the dependence on fixed viewpoints in conventional anomaly detection, a growing line of research has explored pose-agnostic inspection through 3D scene representation and novel-view synthesis. OmniposeAD [167] pioneers this direction by combining coarse-to-fine pose estimation [101], [168] with neural radiance field based reference-image synthesis [169], [170], enabling anomaly detection from arbitrary unknown viewpoints. To improve the efficiency of this paradigm, SplatPose [171] and SplatPose+ [172] replace NeRF-style representations with 3D Gaussian Splatting [173], with the latter further incorporating Structure from Motion [174] to support real-time inference. Extending pose-agnostic detection to more realistic settings, PIAD [175] and R3-PAD [176] further address illumination inconsistency by combining 3D Gaussian Splatting with Retinex-based decomposition or alignment [177], thereby improving robustness under varying lighting and viewpoint conditions. These methods indicate that pose-agnostic anomaly detection is evolving from viewpoint-invariant inspection toward more general scene-consistent anomaly detection that jointly accounts for geometric variation, rendering efficiency, and illumination robustness.

## V. Datasets and Benchmark

### A. Evaluation Metrics

Table I presents the commonly used metrics in 3D anomaly detection. The first column denotes the name of the metric, the

TABLE I
EVALUATION METRICS FOR 3D ANOMALY DETECTION

| Metric | Formula | Remarks | Usage |
|---|---|---|---|
| Precision (P) ↑ | $P = TP/(TP + FP)$ | True positive (TP), false positive (FP) | Classification |
| Recall (R) ↑ | $R = TP/(TP + FN)$ | False negative (FN) | |
| Accuracy (Acc) ↑ | $Acc = (TP + TN)/(TP + TN + FP + FN)$ | | |
| True positive rate (TPR) ↑ | $TPR = TP/(TP + FN)$ | | |
| False positive rate (FPR) ↓ | $FPR = FP/(FP + TN)$ | True negative (TN) | |
| AU-ROC (P-AUROC/O-AUROC) ↑ | $\int_0^1 (TPR)\,\mathrm{d}(FPR)$ | | Classification Detection, Localization |
| AU-PR (P-AUPR/O-AUPR) ↑ | $\int_0^1 (P)\,\mathrm{d}(R)$ | | |
| Per-region overlap (PRO)↑ | $PRO = \frac{1}{N}\sum_i \sum_k \frac{P_i \cap C_{i,k}}{C_{i,k}}$ | Total ground truth number (N) Predicted abnormal pixels (P) Defect ground truth regions (C) | Localization |
| Saturated PRO (sPRO)↑ | $sPRO(P) = \frac{1}{m}\sum_{i=1}^{m} \min(\frac{A_i \cap P}{s_i}, 1)$ | Total ground truth number (m) Predicted abnormal pixels (P) Defect ground truth regions (A) Corresponding saturation thresholds (s) | Localization |
| F1 score ↑ | $F1 = 2(P \times R)/(P + R)$ | Prediction (H), ground truth (G) | Classification Detection, Localization |
| Intersection over union (IoU) ↑ | $IoU = (H \cap G)/(H \cup G)$ | | |
| Memory ↓ | – | Unit: GB | Consumption Efficiency |
| Frames per second (FPS) ↑ | $FPS = 1/T_{\text{inf}}$ | Unit: frames/second | |

TABLE II
COMPARISON OF DATASETS FOR 3D ANOMALY DETECTION

| Dataset | Year | Class | Normal | Abnormal | Total | Real or synthetic | Modality | Defect types |
|---|---|---|---|---|---|---|---|---|
| MVTec 3D-AD [178] | 2022 | 10 | 3 199 | 948 | 4 147 | Real | RGB-D | – |
| Eyecandies [179] | 2022 | 10 | 13 000 | 2 000 | 15 000 | Synthetic | RGB-D | 4 |
| PD-REAL [180] | 2023 | 15 | – | – | 3 500+ | Synthetic | RGB-D | 6 |
| PAD [167] | 2023 | 20 | 5 231 | 4 902 | 10 133+ | Real+Synthetic | RGB (Multi-view) | 3 |
| Real3D-AD [96] | 2023 | 12 | 652 | 602 | 1 254 | Real | Point cloud | 3 |
| Anomaly-ShapeNet [123] | 2024 | 50 | 888 | 712 | 1 600 | Synthetic | Point cloud | 6 |
| RealIAD [181] | 2024 | 30 | 99,721 | 51,329 | 151,050 | Real | RGB (Multi-view) | 8 |
| 3D-ADAM [182] | 2025 | 28 | – | – | 14 120 | Real | RGB-D | 12 |
| SiM3D [183] | 2025 | 8 | 173 | 160 | 333 | Real | RGB-D (Multi-view) | 3 |
| PIAD [175] | 2025 | 30 | 7 102 | 4 166 | 11 268 | Real+Synthetic | Multi-view RGB | 6 |
| MiniShift [184] | 2025 | 12 | 1257 | 1200 | 2 577 | Synthetic | Point cloud | 4 |
| MulSen-AD [78] | 2025 | 15 | 1541 | 494 | 2035 | Real | Super-modal | 14 |
| Real-IAD $D^3$ [185] | 2025 | 20 | 5 000 | 3 450 | 8 450 | Real | RGB (Multi-view) | 69 |
| Real-IAD Variety [186] | 2025 | 160 | 39905 | 159055 | 198960 | Real | Super-modal (Multi-view) | 23 |
| CPS3D-AD [187] | 2025 | 20 | 300 | 1000 | 1300 | Real | Point cloud | 6 |
| IEC3D-AD [188] | 2025 | 15 | 1 500 | 900 | 2 400 | Real | Point cloud | 5 |

TABLE III
MULTI-CLASS ANOMALY DETECTION FOR VARIOUS REAL INDUSTRIAL 3D DATA

| Domain | Object class | Defect class | Dataset |
|---|---|---|---|
| Industrial Components | TScrew HexagonalBolt JointingStud HoleRetainingRing PullingNail HexagonalNut | Bump, Dent, Scratch, Scar, Deformation | IEC3D-AD |
| Integrated Circuits | Class-1 Class-2 Class-3 Class-4 | Crack, Warping, Depression, Foreign Object | CPS3D-AD |
| Toy | Duck Chicken Seahorse Starfish Fish Piggy | Crack, Hole, Squeeze, Scratch, Open | Real3D-AD & Mulsen-AD |
| Daily Necessities | Tush brush Zipper Cotton Capsule | Broken Outside, Foreign Body, Substandard, Broken Intside | Mulsen-AD |

second column provides the detailed definition of each metric, and the third column indicates the specific performance aspects of anomaly detection that the metric reflects. The most commonly used metrics are object-level and pixel-level AU-ROC, but they are highly susceptible to the imbalance between positive and negative samples [8], [189], [190]. Additional metrics such as AU-PR, PRO, F1 score, IoU, and sPRO have been introduced to address this limitation. Furthermore, due to the massive scale of point clouds, FPS and memory consumption are also important lightweight evaluation metrics that assess whether a model can be deployed efficiently. However, existing metrics face challenges when comprehensively evaluating the performance of 3D anomaly detection. We emphasise the need to assign unequal weights to normal and anomalous regions, placing greater emphasis on assessing the model's ability to identify anomalous areas while simultaneously evaluating the impact of normal region scores on the detection of anomalies.

**Open-set setting.** Testing categories or anomaly patterns may not be fully covered during training. Therefore, the model must generalize to unseen categories and anomaly types, making this setting closer to real-world deployment.

TABLE IV
O-AUROC RESULTS ON THE PROPOSED BENCHMARK.

| O-AUROC | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | **PatchCore** | | | **BTF** | | **M3DM** | | **Reg3D-AD** | **ISMP** | **R3D-AD** | **PO3AD** | **MC3D-AD** |
| **Datasets** | **FPFH** | **FPFH+Raw** | **PointMAE** | **Raw** | **FPFH** | **PointMAE** | **PointBert** | **PointMAE** | **PointMAE+FPFH** | **Coordinates** | **Coordinates** | **PointMAE** |
| square gasket | 0.599 | 0.664 | 0.536 | 0.782 | 0.453 | 0.882 | **0.917** | 0.580 | 0.489 | 0.755 | 0.776 | 0.692 |
| cotter pin | 0.496 | 0.538 | 0.458 | **0.889** | 0.488 | 0.881 | 0.758 | 0.540 | 0.443 | 0.657 | 0.363 | 0.800 |
| caliber nut | 0.492 | 0.508 | 0.530 | 0.913 | 0.439 | **0.931** | 0.919 | 0.524 | 0.506 | 0.623 | 0.303 | 0.733 |
| threaded nail | 0.547 | 0.514 | 0.498 | **0.976** | 0.464 | 0.952 | 0.916 | 0.548 | 0.507 | 0.804 | 0.357 | 0.783 |
| L screw | 0.507 | 0.480 | 0.472 | **0.953** | 0.499 | 0.951 | 0.884 | 0.550 | 0.592 | 0.624 | 0.407 | 0.745 |
| retaining ring | 0.495 | 0.625 | 0.594 | 0.995 | 0.518 | **1.000** | **1.000** | 0.646 | 0.660 | 0.703 | 0.436 | 0.708 |
| internal gear washer | 0.524 | 0.590 | 0.674 | 0.980 | 0.631 | 0.996 | **0.999** | 0.727 | 0.530 | 0.812 | 0.466 | 0.724 |
| double ear gasket | 0.501 | 0.606 | 0.385 | **0.991** | 0.428 | 0.898 | 0.877 | 0.512 | 0.610 | 0.811 | 0.762 | 0.573 |
| four sided plug | 0.568 | 0.543 | 0.483 | 0.846 | 0.424 | **0.946** | 0.847 | 0.466 | 0.486 | 0.668 | 0.590 | 0.746 |
| S hook | 0.469 | 0.541 | 0.467 | **0.994** | 0.516 | 0.924 | 0.841 | 0.421 | 0.594 | 0.619 | 0.653 | 0.856 |
| external gear washer | 0.509 | 0.607 | 0.466 | 0.968 | 0.487 | 0.998 | **0.999** | 0.559 | 0.461 | 0.638 | 0.570 | 0.774 |
| hexagon socket head plug | 0.514 | 0.533 | 0.640 | 0.801 | 0.546 | **0.844** | 0.828 | 0.689 | 0.614 | 0.656 | 0.510 | 0.831 |
| self tapping screw | 0.567 | 0.482 | 0.449 | **0.969** | 0.495 | 0.894 | 0.871 | 0.418 | 0.456 | 0.664 | 0.406 | 0.763 |
| round nut | 0.538 | 0.541 | 0.516 | **1.000** | 0.565 | **1.000** | 0.999 | 0.598 | 0.599 | 0.780 | 0.417 | 0.767 |
| medium sized gasket | 0.442 | 0.575 | 0.449 | 0.916 | 0.457 | **0.994** | 0.982 | 0.512 | 0.493 | 0.731 | 0.429 | 0.909 |
| Class_1 | 0.508 | 0.432 | 0.431 | 0.651 | 0.474 | 0.805 | **0.834** | 0.451 | 0.404 | 0.575 | 0.495 | 0.731 |
| Class_2 | 0.493 | 0.505 | 0.580 | 0.551 | 0.375 | **0.795** | 0.783 | 0.473 | 0.549 | 0.554 | 0.732 | 0.569 |
| Class_3 | 0.665 | 0.617 | 0.590 | 0.473 | 0.388 | 0.367 | 0.686 | 0.766 | 0.644 | **0.798** | 0.681 | 0.543 |
| Class_4 | 0.507 | 0.600 | 0.604 | 0.645 | 0.514 | 0.760 | **0.782** | 0.586 | 0.469 | 0.587 | 0.419 | 0.710 |
| airplane | **0.882** | 0.848 | 0.726 | 0.520 | 0.653 | 0.333 | 0.407 | 0.716 | 0.858 | 0.772 | 0.804 | 0.850 |
| car | 0.590 | **0.777** | 0.498 | 0.560 | 0.614 | 0.376 | 0.506 | 0.697 | 0.731 | 0.696 | 0.654 | 0.749 |
| candybar | 0.565 | 0.626 | 0.585 | 0.462 | 0.517 | 0.363 | 0.442 | 0.685 | **0.852** | 0.713 | 0.785 | 0.830 |
| chicken | 0.837 | **0.853** | 0.827 | 0.432 | 0.417 | 0.383 | 0.673 | 0.852 | 0.714 | 0.714 | 0.686 | 0.715 |
| diamond | 0.574 | 0.784 | 0.783 | 0.545 | 0.593 | 0.568 | 0.627 | 0.900 | 0.948 | 0.685 | 0.801 | **0.955** |
| duck | 0.546 | 0.628 | 0.489 | 0.784 | 0.498 | 0.419 | 0.466 | 0.584 | 0.712 | **0.909** | 0.820 | 0.831 |
| fish | 0.675 | 0.837 | 0.630 | 0.549 | 0.318 | 0.512 | 0.556 | 0.915 | **0.945** | 0.692 | 0.859 | 0.865 |
| gemstone | 0.370 | 0.359 | 0.374 | 0.648 | 0.516 | 0.491 | 0.617 | 0.417 | 0.468 | 0.665 | **0.693** | 0.560 |
| seahorse | 0.505 | 0.767 | 0.539 | **0.779** | 0.627 | 0.090 | 0.494 | 0.762 | 0.729 | 0.720 | 0.756 | 0.716 |
| shell | 0.589 | 0.663 | 0.501 | 0.754 | 0.369 | 0.631 | 0.577 | 0.583 | 0.623 | **0.840** | 0.800 | 0.803 |
| starfish | 0.441 | 0.471 | 0.519 | 0.575 | 0.459 | 0.490 | 0.528 | 0.506 | 0.660 | 0.701 | 0.758 | **0.766** |
| toffees | 0.541 | 0.570 | 0.663 | 0.630 | 0.442 | 0.502 | 0.562 | 0.827 | **0.842** | 0.703 | 0.771 | 0.738 |
| Mean | 0.550 | 0.603 | 0.547 | **0.759** | 0.490 | 0.709 | 0.748 | 0.613 | 0.619 | 0.705 | 0.611 | 0.753 |

TABLE V
P-AUROC RESULTS ON THE PROPOSED BENCHMARK.

| P-AUROC | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | **PatchCore** | | | **BTF** | | **M3DM** | | **Reg3D-AD** | **ISMP** | **R3D-AD** | **PO3AD** | **MC3D-AD** |
| **Datasets** | **FPFH** | **FPFH+Raw** | **PointMAE** | **Raw** | **FPFH** | **PointMAE** | **PointBert** | **PointMAE** | **PointMAE+FPFH** | **Coordinates** | **Coordinates** | **PointMAE** |
| square gasket | 0.476 | 0.492 | 0.403 | 0.467 | 0.492 | 0.153 | 0.591 | 0.501 | 0.619 | 0.195 | 0.629 | **0.700** |
| cotter pin | 0.609 | 0.523 | 0.488 | 0.618 | 0.489 | 0.251 | 0.190 | 0.580 | 0.429 | 0.270 | 0.535 | **0.728** |
| caliber nut | **0.576** | 0.513 | 0.505 | 0.575 | 0.519 | 0.566 | **0.576** | 0.521 | 0.446 | 0.225 | 0.491 | 0.555 |
| threaded nail | 0.585 | 0.533 | 0.474 | 0.676 | 0.454 | **0.718** | 0.711 | 0.512 | 0.479 | 0.241 | 0.564 | 0.708 |
| L screw | 0.507 | 0.533 | 0.446 | **0.767** | 0.499 | 0.683 | 0.652 | 0.485 | 0.524 | 0.184 | 0.560 | 0.630 |
| retaining ring | 0.540 | 0.520 | 0.499 | **0.724** | 0.569 | 0.454 | 0.577 | 0.531 | 0.630 | 0.165 | 0.466 | 0.557 |
| internal gear washer | 0.545 | 0.610 | 0.562 | 0.696 | 0.501 | **0.725** | 0.704 | 0.601 | 0.482 | 0.258 | 0.566 | 0.588 |
| double ear gasket | 0.476 | 0.533 | 0.426 | 0.723 | 0.513 | **0.753** | 0.752 | 0.524 | 0.518 | 0.202 | 0.584 | 0.721 |
| four sided plug | 0.532 | 0.507 | 0.453 | **0.649** | 0.427 | 0.358 | 0.192 | 0.537 | 0.541 | 0.326 | 0.485 | 0.624 |
| S hook | 0.519 | 0.513 | 0.507 | 0.679 | 0.551 | 0.887 | **0.896** | 0.510 | 0.509 | 0.232 | 0.535 | 0.778 |
| external gear washer | 0.582 | 0.634 | 0.639 | 0.701 | 0.604 | 0.689 | 0.623 | 0.594 | 0.639 | 0.234 | 0.566 | **0.824** |
| hexagon socket head plug | 0.583 | 0.520 | 0.492 | **0.671** | 0.531 | 0.551 | 0.527 | 0.572 | 0.488 | 0.253 | 0.515 | 0.638 |
| self tapping screw | 0.521 | 0.524 | 0.612 | 0.637 | 0.531 | 0.563 | 0.545 | 0.573 | 0.409 | 0.175 | 0.506 | **0.706** |
| round nut | 0.555 | 0.490 | 0.566 | 0.727 | 0.594 | 0.739 | 0.724 | 0.505 | 0.425 | 0.163 | 0.583 | **0.858** |
| medium sized gasket | 0.546 | 0.527 | 0.537 | 0.636 | 0.397 | 0.941 | 0.934 | 0.534 | 0.515 | 0.201 | 0.545 | **0.943** |
| Class_1 | 0.696 | 0.698 | 0.703 | 0.673 | 0.766 | 0.931 | **0.941** | 0.692 | 0.491 | 0.325 | 0.614 | 0.804 |
| Class_2 | 0.724 | 0.720 | 0.666 | 0.564 | 0.650 | 0.916 | **0.932** | 0.647 | 0.416 | 0.273 | 0.395 | 0.552 |
| Class_3 | 0.733 | 0.657 | 0.581 | 0.554 | 0.741 | 0.911 | **0.933** | 0.553 | 0.562 | 0.378 | 0.392 | 0.625 |
| Class_4 | 0.688 | 0.671 | 0.679 | 0.628 | 0.675 | **0.946** | 0.934 | 0.670 | 0.513 | 0.418 | 0.588 | 0.716 |
| airplane | 0.471 | 0.556 | 0.579 | 0.564 | 0.839 | **0.857** | 0.523 | 0.631 | 0.753 | 0.594 | 0.542 | 0.628 |
| car | 0.643 | 0.740 | 0.610 | 0.647 | 0.917 | **0.924** | 0.593 | 0.718 | 0.836 | 0.557 | 0.599 | 0.819 |
| candybar | 0.637 | 0.749 | 0.635 | 0.735 | 0.888 | **0.933** | 0.682 | 0.724 | 0.907 | 0.593 | 0.680 | 0.910 |
| chicken | 0.618 | 0.558 | 0.683 | 0.608 | 0.832 | **0.848** | 0.790 | 0.676 | 0.798 | 0.620 | 0.670 | 0.640 |
| diamond | 0.760 | 0.854 | 0.776 | 0.563 | 0.816 | 0.875 | 0.594 | 0.835 | 0.926 | 0.555 | 0.604 | **0.942** |
| duck | 0.430 | 0.658 | 0.439 | 0.601 | 0.750 | 0.837 | 0.668 | 0.503 | **0.876** | 0.635 | 0.657 | 0.822 |
| fish | 0.464 | 0.781 | 0.714 | 0.514 | 0.920 | 0.888 | 0.589 | 0.826 | 0.886 | 0.573 | 0.586 | **0.932** |
| gemstone | 0.830 | 0.539 | 0.514 | 0.597 | 0.799 | **0.861** | 0.646 | 0.545 | 0.857 | 0.668 | 0.656 | 0.458 |
| seahorse | 0.544 | 0.808 | 0.660 | 0.520 | 0.943 | **0.955** | 0.574 | 0.817 | 0.813 | 0.562 | 0.552 | 0.659 |
| shell | 0.596 | 0.753 | 0.725 | 0.489 | 0.741 | 0.827 | 0.732 | 0.811 | **0.839** | 0.578 | 0.729 | 0.778 |
| starfish | 0.522 | 0.613 | 0.641 | 0.392 | 0.917 | **0.929** | 0.563 | 0.617 | 0.641 | 0.608 | 0.538 | 0.690 |
| toffees | 0.411 | 0.549 | 0.727 | 0.623 | 0.762 | 0.883 | 0.677 | 0.759 | 0.895 | 0.568 | 0.684 | **0.934** |
| Mean | 0.578 | 0.609 | 0.579 | 0.620 | 0.665 | **0.753** | 0.663 | 0.616 | 0.634 | 0.382 | 0.568 | 0.725 |

### *B. Datasets*

The acquisition of 3D data is extremely challenging, especially as obtaining high-precision point clouds requires substantial manual labor. We summarize all known datasets in Table II and present some negative samples in Fig. 7, demonstrating the continuous growth of high-quality data in this field and a gradual shift toward application-oriented real-world data. Among these, one challenge is the insufficient positive training data and lack of supervised negative examples. Due to the unclear definition of anomalies, models typically struggle to obtain a clear optimization direction. A combination of real anomaly data and simulated anomalies may bridge the gap between models and real-world applications. We anticipate acquiring more high-quality data with clear supervisory signals from actual production lines in the future.

### *C. Benchmark: Multi-class Anomaly Detection for Various Real Industrial 3D Data*

To comprehensively and effectively evaluate the value of existing models and identify future directions for improve-

TABLE VI
O-AUPR RESULTS ON THE PROPOSED BENCHMARK

| O-AUPR | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | **PatchCore** | | | **BTF** | | **M3DM** | | **Reg3D-AD** | **ISMP** | **R3D-AD** | **PO3AD** | **MC3D-AD** |
| **Datasets** | **FPFH** | **FPFH+Raw** | **PointMAE** | **Raw** | **FPFH** | **PointMAE** | **PointBert** | **PointMAE** | **PointMAE+FPFH** | **Coordinates** | **Coordinates** | **PointMAE** |
| square gasket | 0.605 | 0.650 | 0.581 | 0.819 | 0.479 | 0.928 | **0.941** | 0.610 | 0.496 | 0.669 | 0.811 | 0.692 |
| cotter pin | 0.474 | 0.595 | 0.484 | **0.915** | 0.505 | 0.910 | 0.811 | 0.554 | 0.467 | 0.591 | 0.451 | 0.800 |
| caliber nut | 0.494 | 0.526 | 0.513 | 0.933 | 0.450 | **0.950** | 0.947 | 0.516 | 0.523 | 0.551 | 0.419 | 0.733 |
| threaded nail | 0.568 | 0.489 | 0.532 | **0.981** | 0.512 | 0.963 | 0.932 | 0.565 | 0.549 | 0.737 | 0.427 | 0.783 |
| L screw | 0.497 | 0.497 | 0.504 | **0.971** | 0.509 | 0.959 | 0.890 | 0.549 | 0.565 | 0.557 | 0.494 | 0.745 |
| retaining ring | 0.520 | 0.596 | 0.559 | 0.995 | 0.522 | **1.000** | **1.000** | 0.628 | 0.679 | 0.632 | 0.578 | 0.736 |
| internal gear washer | 0.534 | 0.599 | 0.694 | 0.984 | 0.639 | 0.997 | **0.999** | 0.728 | 0.535 | 0.711 | 0.575 | 0.772 |
| double ear gasket | 0.538 | 0.640 | 0.457 | **0.992** | 0.480 | 0.906 | 0.915 | 0.535 | 0.583 | 0.709 | 0.839 | 0.576 |
| four sided plug | 0.538 | 0.500 | 0.510 | 0.868 | 0.448 | **0.963** | 0.872 | 0.491 | 0.484 | 0.573 | 0.604 | 0.746 |
| S hook | 0.479 | 0.530 | 0.502 | **0.995** | 0.549 | 0.933 | 0.850 | 0.441 | 0.611 | 0.555 | 0.625 | 0.856 |
| external gear washer | 0.508 | 0.634 | 0.522 | 0.976 | 0.529 | 0.998 | **0.999** | 0.638 | 0.488 | 0.585 | 0.692 | 0.774 |
| hexagon socket head plug | 0.519 | 0.587 | 0.641 | 0.790 | 0.517 | **0.842** | 0.834 | 0.672 | 0.577 | 0.593 | 0.625 | 0.831 |
| self tapping screw | 0.582 | 0.505 | 0.455 | **0.974** | 0.538 | 0.852 | 0.867 | 0.465 | 0.504 | 0.610 | 0.535 | 0.763 |
| round nut | 0.519 | 0.586 | 0.521 | **1.000** | 0.554 | **1.000** | 0.999 | 0.604 | 0.592 | 0.819 | 0.484 | 0.767 |
| medium sized gasket | 0.492 | 0.603 | 0.476 | 0.929 | 0.480 | **0.994** | 0.988 | 0.543 | 0.534 | 0.631 | 0.557 | 0.909 |
| Class_1 | 0.955 | 0.940 | 0.942 | 0.972 | 0.952 | 0.985 | **0.990** | 0.940 | 0.936 | 0.962 | 0.951 | 0.981 |
| Class_2 | 0.904 | 0.902 | 0.909 | 0.913 | 0.870 | **0.974** | 0.969 | 0.868 | 0.904 | 0.917 | 0.941 | 0.936 |
| Class_3 | 0.964 | 0.924 | 0.941 | 0.897 | 0.904 | 0.885 | 0.943 | 0.966 | 0.961 | **0.975** | 0.960 | 0.917 |
| Class_4 | 0.940 | 0.959 | 0.952 | 0.963 | 0.935 | 0.979 | **0.982** | 0.954 | 0.936 | 0.957 | 0.918 | 0.970 |
| airplane | 0.852 | 0.807 | 0.747 | 0.505 | 0.534 | 0.528 | 0.497 | 0.703 | 0.566 | 0.514 | 0.531 | **0.890** |
| car | 0.611 | **0.766** | 0.555 | 0.523 | 0.592 | 0.582 | 0.517 | 0.753 | 0.666 | 0.508 | 0.525 | 0.700 |
| candybar | 0.553 | 0.611 | 0.576 | 0.490 | 0.695 | 0.690 | 0.480 | 0.824 | 0.663 | 0.519 | 0.541 | **0.854** |
| chicken | 0.872 | **0.885** | 0.864 | 0.464 | 0.655 | 0.650 | 0.716 | 0.884 | 0.607 | 0.569 | 0.595 | 0.724 |
| diamond | 0.569 | 0.767 | 0.801 | 0.535 | 0.591 | 0.601 | 0.661 | 0.884 | 0.900 | 0.682 | 0.717 | **0.919** |
| duck | 0.506 | 0.560 | 0.488 | **0.760** | 0.661 | 0.650 | 0.569 | 0.588 | 0.488 | 0.569 | 0.612 | 0.613 |
| fish | 0.642 | 0.844 | 0.720 | 0.633 | 0.580 | 0.595 | 0.628 | **0.939** | 0.825 | 0.634 | 0.736 | 0.878 |
| gemstone | 0.411 | 0.411 | 0.444 | 0.598 | **0.692** | 0.685 | 0.628 | 0.454 | 0.567 | 0.617 | 0.690 | 0.465 |
| seahorse | 0.508 | 0.763 | 0.546 | **0.793** | 0.595 | 0.585 | 0.491 | 0.787 | 0.770 | 0.538 | 0.579 | 0.744 |
| shell | 0.573 | 0.553 | 0.590 | 0.751 | 0.720 | 0.710 | 0.638 | 0.646 | 0.521 | 0.605 | 0.539 | **0.860** |
| starfish | 0.491 | 0.473 | 0.561 | 0.579 | 0.570 | 0.580 | 0.573 | 0.491 | 0.465 | 0.544 | **0.709** | 0.638 |
| toffees | 0.506 | 0.559 | 0.708 | 0.700 | 0.680 | 0.675 | 0.569 | 0.721 | 0.555 | 0.527 | 0.697 | **0.799** |
| Mean | 0.604 | 0.654 | 0.622 | 0.813 | 0.611 | **0.824** | 0.797 | 0.675 | 0.630 | 0.650 | 0.644 | 0.786 |

TABLE VII
P-AUPR RESULTS ON THE PROPOSED BENCHMARK

| P-AUPR | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | **PatchCore** | | | **BTF** | | **M3DM** | | **Reg3D-AD** | **ISMP** | **R3D-AD** | **PO3AD** | **MC3D-AD** |
| **Datasets** | **FPFH** | **FPFH+Raw** | **PointMAE** | **Raw** | **FPFH** | **PointMAE** | **PointBert** | **PointMAE** | **PointMAE+FPFH** | **Coordinates** | **Coordinates** | **PointMAE** |
| square gasket | 0.006 | 0.006 | 0.004 | 0.006 | 0.008 | 0.008 | 0.014 | 0.006 | 0.009 | 0.004 | **0.051** | 0.018 |
| cotter pin | 0.011 | 0.007 | 0.008 | 0.011 | 0.008 | 0.009 | 0.009 | 0.010 | 0.006 | 0.004 | 0.008 | **0.182** |
| caliber nut | 0.005 | 0.004 | 0.003 | 0.004 | 0.004 | 0.001 | 0.001 | 0.003 | 0.003 | 0.002 | 0.003 | **0.066** |
| threaded nail | 0.010 | 0.008 | 0.016 | 0.013 | 0.006 | 0.003 | 0.003 | 0.024 | 0.007 | 0.005 | 0.009 | **0.216** |
| L screw | 0.007 | 0.006 | 0.005 | 0.014 | 0.005 | 0.001 | 0.001 | 0.005 | 0.006 | 0.003 | 0.007 | **0.098** |
| retaining ring | 0.005 | 0.005 | 0.005 | 0.010 | 0.007 | 0.013 | 0.014 | 0.005 | 0.011 | 0.003 | 0.004 | **0.016** |
| internal gear washer | 0.007 | 0.019 | 0.024 | 0.011 | 0.007 | 0.001 | 0.001 | 0.017 | 0.006 | 0.004 | 0.008 | **0.120** |
| double ear gasket | 0.005 | 0.008 | 0.005 | 0.010 | 0.006 | 0.029 | 0.022 | 0.006 | 0.006 | 0.003 | 0.009 | **0.048** |
| four sided plug | 0.003 | 0.003 | 0.002 | 0.006 | 0.003 | 0.011 | 0.006 | 0.003 | 0.003 | 0.002 | 0.003 | **0.064** |
| S hook | 0.008 | 0.007 | 0.007 | 0.013 | 0.009 | 0.065 | 0.071 | 0.006 | 0.007 | 0.004 | 0.008 | **0.129** |
| external gear washer | 0.007 | 0.041 | 0.066 | 0.011 | 0.009 | 0.035 | 0.025 | 0.012 | 0.008 | 0.003 | 0.006 | **0.285** |
| hexagon socket head plug | 0.004 | 0.003 | 0.003 | 0.007 | 0.004 | 0.020 | 0.016 | 0.004 | 0.003 | 0.002 | 0.003 | **0.047** |
| self tapping screw | 0.014 | 0.014 | 0.026 | 0.026 | 0.017 | 0.016 | 0.012 | 0.023 | 0.010 | 0.008 | 0.013 | **0.082** |
| round nut | 0.007 | 0.006 | 0.010 | 0.012 | 0.010 | 0.002 | 0.002 | 0.006 | 0.005 | 0.004 | 0.009 | **0.077** |
| medium sized gasket | 0.007 | 0.008 | 0.012 | 0.012 | 0.005 | 0.177 | 0.095 | 0.006 | 0.007 | 0.004 | 0.011 | **0.381** |
| Class_1 | 0.024 | 0.024 | 0.024 | 0.032 | 0.032 | 0.108 | **0.280** | 0.023 | 0.012 | 0.008 | 0.028 | 0.229 |
| Class_2 | 0.023 | 0.021 | 0.014 | 0.016 | 0.015 | 0.108 | **0.376** | 0.013 | 0.007 | 0.006 | 0.008 | 0.020 |
| Class_3 | 0.030 | 0.020 | 0.015 | 0.017 | 0.030 | 0.097 | **0.309** | 0.014 | 0.018 | 0.010 | 0.016 | 0.057 |
| Class_4 | 0.007 | 0.006 | 0.006 | 0.050 | 0.006 | 0.188 | **0.213** | 0.006 | 0.004 | 0.003 | 0.005 | 0.034 |
| airplane | 0.027 | 0.016 | 0.016 | 0.012 | 0.007 | 0.006 | 0.007 | 0.017 | 0.008 | 0.008 | 0.008 | **0.046** |
| car | 0.034 | 0.160 | 0.069 | 0.014 | 0.010 | 0.009 | 0.017 | 0.135 | 0.027 | 0.013 | 0.019 | **0.163** |
| candybar | 0.142 | 0.092 | 0.020 | 0.025 | 0.018 | 0.014 | 0.016 | 0.109 | 0.015 | 0.016 | 0.016 | **0.491** |
| chicken | 0.040 | 0.045 | 0.052 | 0.049 | 0.046 | 0.052 | **0.377** | 0.044 | 0.032 | 0.048 | 0.052 | 0.049 |
| diamond | 0.273 | 0.363 | 0.107 | 0.032 | 0.108 | 0.100 | 0.038 | 0.191 | 0.064 | 0.038 | 0.043 | **0.582** |
| duck | **0.055** | 0.034 | 0.008 | 0.020 | 0.014 | 0.009 | 0.011 | 0.010 | 0.010 | 0.011 | 0.011 | 0.032 |
| fish | 0.052 | 0.266 | 0.201 | 0.017 | 0.098 | 0.107 | 0.039 | 0.437 | 0.033 | 0.035 | 0.084 | **0.532** |
| gemstone | **0.093** | 0.066 | 0.008 | 0.014 | 0.015 | 0.020 | 0.017 | 0.016 | 0.013 | 0.018 | 0.027 | 0.006 |
| seahorse | 0.031 | 0.291 | 0.071 | 0.031 | 0.030 | 0.025 | 0.028 | 0.182 | 0.041 | 0.027 | 0.031 | **0.300** |
| shell | 0.031 | 0.049 | 0.043 | 0.011 | 0.003 | 0.005 | 0.021 | 0.065 | 0.022 | 0.020 | 0.021 | **0.344** |
| starfish | 0.037 | 0.035 | 0.046 | 0.017 | 0.038 | 0.045 | 0.040 | 0.039 | 0.021 | 0.037 | 0.041 | **0.102** |
| toffees | 0.040 | 0.055 | 0.055 | 0.016 | 0.025 | 0.020 | 0.018 | 0.067 | 0.013 | 0.022 | 0.032 | **0.414** |
| Mean | 0.034 | 0.054 | 0.031 | 0.017 | 0.019 | 0.042 | 0.068 | 0.049 | 0.014 | 0.012 | 0.019 | **0.169** |

ment, we summarize a benchmark dataset covering common types, including Industrial Equipment Components, Integrated Circuits, Toy, and Daily Necessities, as shown in Table III, to assess the performance of current models across various aspects. Tables IV, V, VI, and VII report the statistical results of current 3D-AD methods on a unified real industrial benchmark. These tables correspond to O-AUROC, P-AUROC, O-AUPR, and P-AUPR. We evaluate several representative approaches. Fig. 8 displays partial visualization maps of our benchmarks. Most of them emphasize sample-level detection, while pixel-level detection and precise localization remain highly challenging. Based on the results, we provide an in-depth analysis and summarize the following observations.

1) Feature-embedding architectures are currently the most effective designs for both sample-level and pixel-level anomaly detection. Methods that adopt handcrafted operators such as FPFH further indicate that low-level geometric representations of point clouds play a critical role in anomaly detection.
2) Group-level feature representations can utilize neigh-

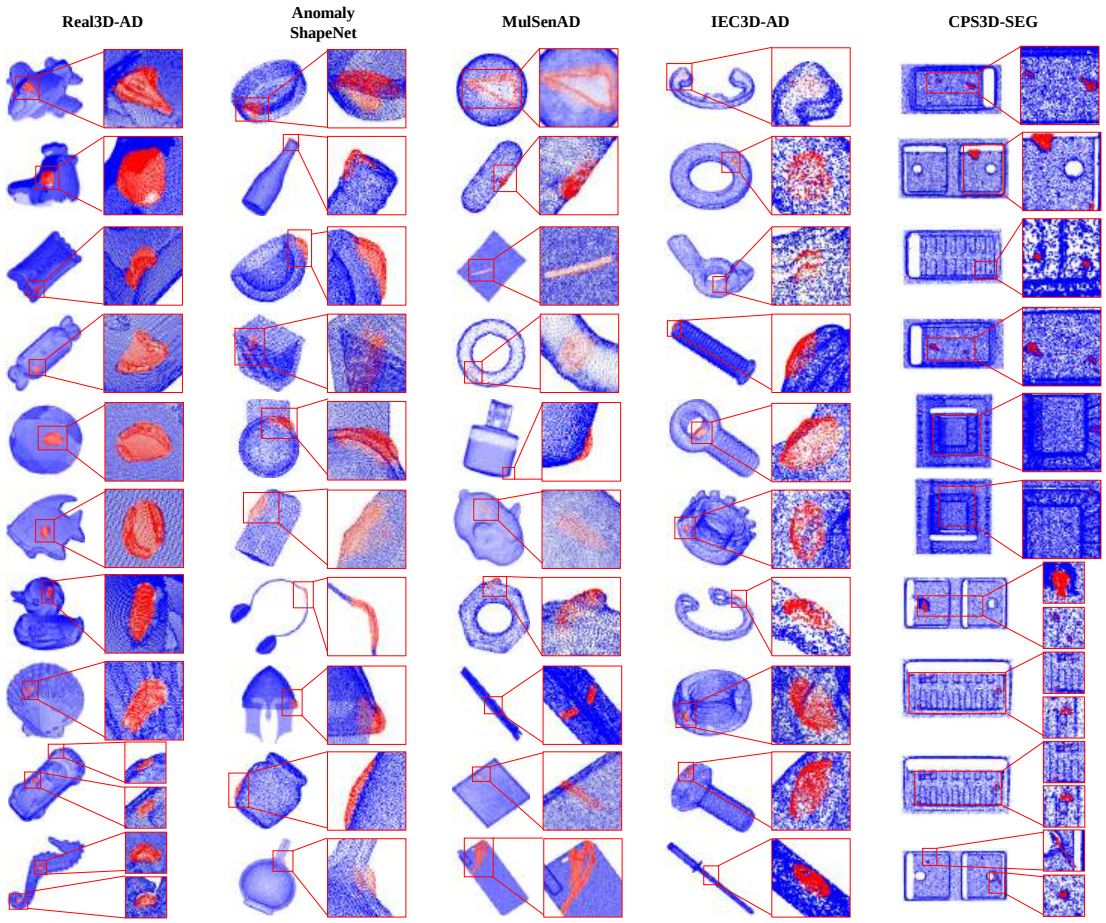



Fig. 7. Partial negative data visualization of different datasets.

borhood information and prior knowledge. However, their relatively low representation density often leads to weaker localization performance.

3) Existing 3D anomaly detection methods lack diverse strategies for generating negative samples. This limitation restricts the optimization direction of many models.
4) AUPR reflects model capability more reliably than AUROC. As shown by the comparison between Tables VI, and VII, the number of positive points in point clouds is much larger than the number of negative points. Although the evaluated methods achieve competitive performance in P-AUROC, they exhibit notable performance degradation when positive and negative samples are assessed equally. MC3D-AD achieves the best performance, suggesting that feature-reconstruction-based methods are more suitable for localization than feature-embedding-based methods.
5) Current 3D-AD methods are still in an early development stage. Precise point-cloud segmentation remains difficult. Existing approaches mainly rely on simple feature extraction followed by secondary processing. Techniques that have proven effective in image anomaly detection, such as flow-based models and diffusion models, have not yet been fully explored in 3D-AD.

## VI. Challenges and Future Work

Attributable to the high-resolution challenges in industrial anomaly detection, certain anomalies may prove difficult to detect, as illustrated in Fig 9. We outline several intriguing future directions as follows:

1) Improving Data Acquisition Efficiency. While blue laser scanners can acquire high-precision point cloud data, the lengthy acquisition process hinders industrial efficiency. Therefore, developing accurate anomaly detection methods adaptable to sparse point clouds or significantly enhancing pipeline acquisition speed represent two crucial directions for advancing the practical deployment of 3D anomaly detection in industrial settings.
2) Improving Model Detection Efficiency. Extracting features from high-precision point clouds, particularly when encoding point-level details, often poses significant efficiency challenges. Developing novel methods that maintain fine-grained representation capabilities while substantially enhancing processing efficiency represents a critical improvement direction for advancing the practical application of this technology.
3) Expanding Training Data. The insufficiency of training data, particularly the scarcity of negative samples, significantly hinders the model's ability to accurately learn normal feature representations. Constructing

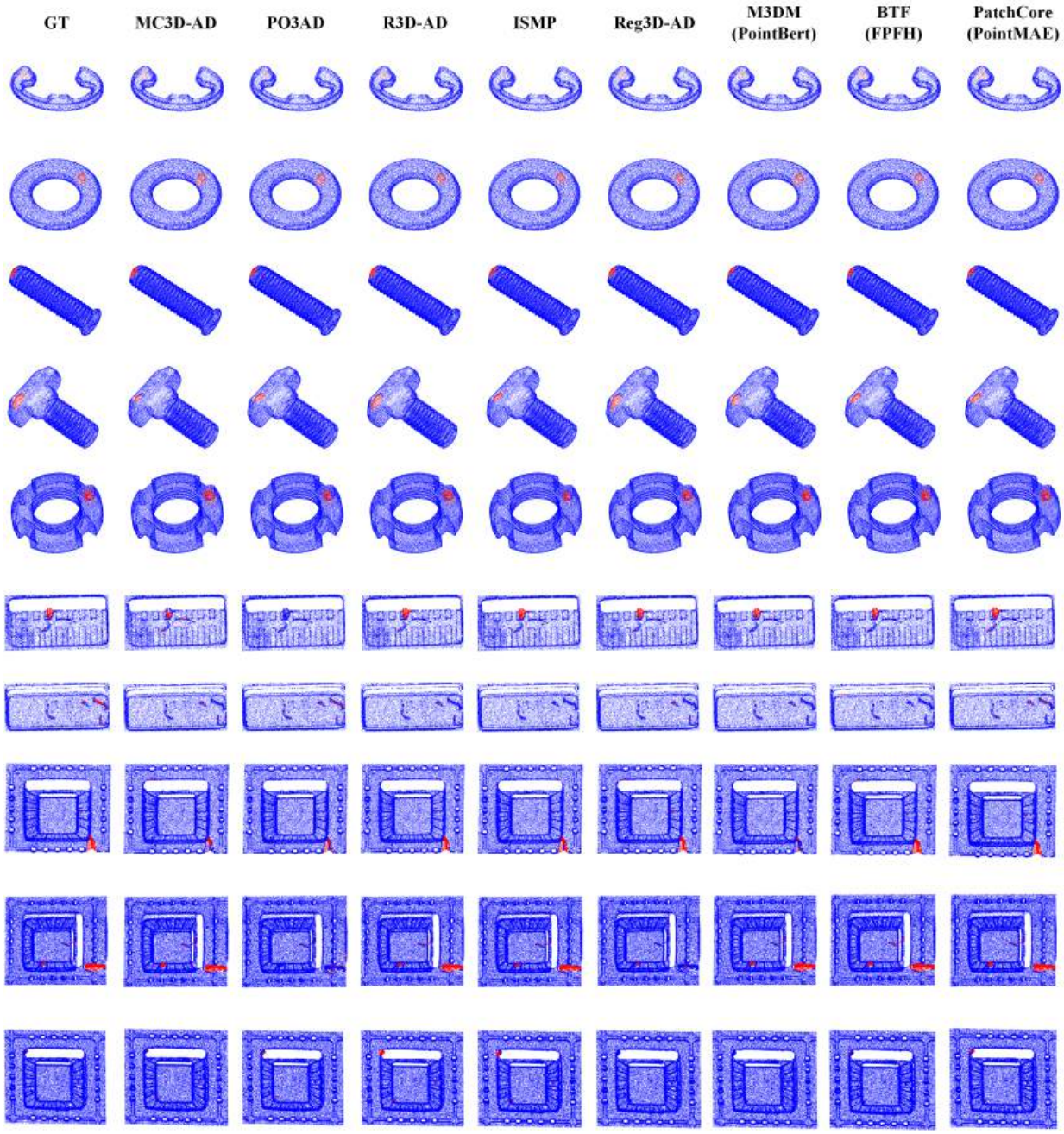



Fig. 8. Partial visualization maps of our benchmarks.

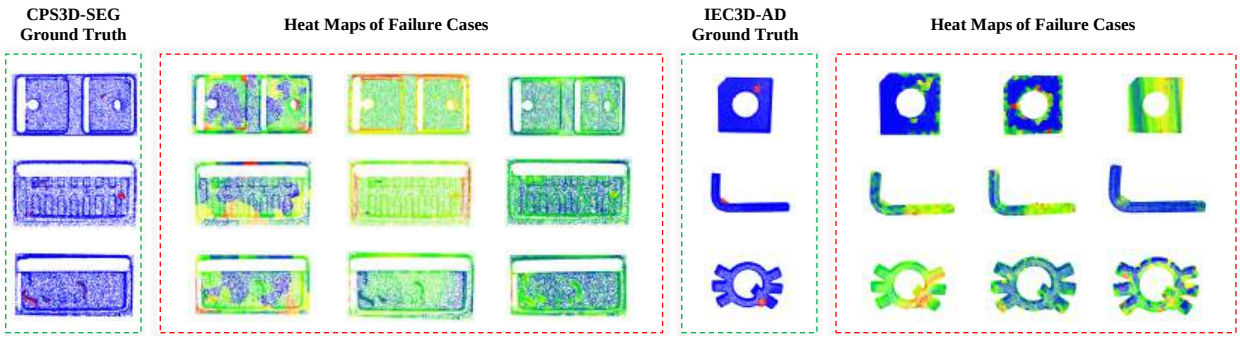



Fig. 9. Some instances of failed detection.

fully-supervised datasets with high-quality annotations presents a highly promising direction to address this challenge.

4) Advancing Anomaly Synthesis Techniques. Current anomaly generation methods often produce samples that significantly deviate from real-world industrial anomaly patterns. There is an urgent need to develop anomaly synthesis techniques that can accurately simulate actual industrial defects, thereby effectively enhancing the model's discrimination capability.
5) Enhancing Model Robustness. Model performance can be significantly compromised by real-world industrial factors such as sensor noise and lighting variations. Implementing advanced approaches like disentangled representation learning offers a promising path to enhance model adaptability and stability in complex industrial environments.
6) Developing Specialized Encoders. Current encoders often lack pretraining on high-resolution point clouds and specific optimization for quality inspection tasks, particularly in capturing fine-grained point-level representations, resulting in suboptimal performance in real industrial applications. Conducting targeted pretraining on high-resolution industrial point cloud data with defect-oriented proxy tasks represents a crucial approach to enhancing model effectiveness in practical quality control settings.
7) Building Foundation Models. Existing models are typically trained independently on datasets with homogeneous distributions, which limits their applicability in real industrial environments. Developing cross-distribution detection capabilities that can adapt to varying data distributions and attributes represents a crucial direction for achieving industrial-grade deployment.
8) Understanding 3D Anomalies with Process Knowledge. Integrating manufacturing process knowledge bases with multimodal large language models enables deeper interpretation of detected anomalies, explaining defect origins and suggesting repair strategies based on domain expertise.
9) Embodied AI for Automated Quality Control. Integrating 3D anomaly detection with embodied AI systems such as vision-language-action models (VLAs) and robotic manipulation enables closed-loop quality control, where robots can autonomously locate, grasp, and remove defective products or perform corrective operations on production lines.

## VII. Conclusion

In this paper, we provide a comprehensive literature review on 3D anomaly detection, systematically investigating multiple dimensions including modality classification, model design, dataset types, evaluation metrics, and comprehensive assessment. We thoroughly analyze the characteristics and applicable scenarios of different modality data, examine the design principles and technical evolution of current mainstream models, summarize the advantages and limitations of existing datasets, and discuss the rationality and effectiveness of evaluation metrics. Furthermore, we conduct an in-depth exploration of future research directions targeting the key challenges faced in the field of 3D anomaly detection, providing valuable insights to advance further development in this domain.

## References


[1] Y. Cao, X. Xu, J. Zhang, Y. Cheng, X. Huang, G. Pang, and W. Shen, "A survey on visual anomaly detection: Challenge, approach, and prospect," 2024.

[2] J. Du, C. Tao, X. Cao, and F. Tsung, "3d vision-based anomaly detection in manufacturing: A survey," *Frontiers of Engineering Management*, vol. 12, no. 2, pp. 343–360, 2025.

[3] A.-S. Moon, K. Lee, and J. Lee, "Survey on 3d anomaly detection regarding defect size: Tradeoff between accuracy and efficiency with future research direction," *IEEE Access*, 2025.

[4] Z. Li, Y. Yan, X. Wang, Y. Ge, and L. Meng, "A survey of deep learning for industrial visual anomaly detection," *Artificial Intelligence Review*, vol. 58, no. 9, p. 279, 2025.

[5] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM computing surveys (CSUR)*, vol. 41, no. 3, pp. 1–58, 2009.

[6] G. Pang, C. Shen, L. Cao, and A. V. D. Hengel, "Deep learning for anomaly detection: A review," *ACM computing surveys (CSUR)*, vol. 54, no. 2, pp. 1–38, 2021.

[7] J. Liu, G. Xie, J. Wang, S. Li, C. Wang, F. Zheng, and Y. Jin, "Deep industrial image anomaly detection: A survey," *Machine Intelligence Research*, vol. 21, no. 1, pp. 104–135, 2024.

[8] G. Xie, J. Wang, J. Liu, J. Lyu, Y. Liu, C. Wang, F. Zheng, and Y. Jin, "Im-iad: Industrial image anomaly detection benchmark in manufacturing," *IEEE Transactions on Cybernetics*, vol. 54, no. 5, pp. 2720–2733, 2024.

[9] W. Luo, Y. Cao, H. Yao, X. Zhang, J. Lou, Y. Cheng, W. Shen, and W. Yu, "Exploring intrinsic normal prototypes within a single image for universal anomaly detection," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 9974–9983, 2025.

[10] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.

[11] M. Rudolph, T. Wehrbein, B. Rosenhahn, and B. Wandt, "Asymmetric student-teacher networks for industrial anomaly detection," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 2592–2602, 2023.

[12] P. Bergmann, K. Batzner, M. Fauser, D. Sattlegger, and C. Steger, "Beyond dents and scratches: Logical constraints in unsupervised anomaly detection and localization," *International Journal of Computer Vision*, vol. 130, no. 4, pp. 947–969, 2022.

[13] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "Uninformed students: Student-teacher anomaly detection with discriminative latent embeddings," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4183–4192, 2020.

[14] L. Dinh, J. Sohl-Dickstein, and S. Bengio, "Density estimation using real nvp," *arXiv preprint arXiv:1605.08803*, 2016.

[15] L. Ardizzone, C. Lüth, J. Kruse, C. Rother, and U. Köthe, "Guided image generation with conditional invertible neural networks," *arXiv preprint arXiv:1907.02392*, 2019.

[16] Y. Cheng, J. Chen, G. Wen, X. Tan, and X. Liu, "Lpfstnet: A lightweight and parameter-free head attention-based student–teacher network for fast 3d industrial anomaly detection," *Neurocomputing*, vol. 623, p. 129408, 2025.

[17] W. Sui, D. Lichau, J. Lefèvre, and H. Phelippeau, "Cross-modal distillation in industrial anomaly detection: Exploring efficient multi-modal iad," *arXiv e-prints*, pp. arXiv–2405, 2024.

[18] F. Zhang, W. Hu, Y. Wang, *et al.*, "Dynamic background-guided asymmetric knowledge distillation network for 3d defect detection," *Journal of Manufacturing Processes*, vol. 160, pp. 185–199, 2026.

[19] C. Wang, J. Chen, and H. Zhang, "Dcrdf-net: A dual-channel reverse-distillation fusion network for 3d industrial anomaly detection," *Sensors (Basel, Switzerland)*, vol. 26, no. 2, p. 412, 2026.

[20] Z. Gu, J. Zhang, L. Liu, X. Chen, J. Peng, Z. Gan, G. Jiang, A. Shu, Y. Wang, and L. Ma, "Rethinking reverse distillation for multi-modal anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 8445–8453, 2024.

[21] H. Deng and X. Li, "Anomaly detection via reverse distillation from one-class embedding," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 9737–9746, 2022.

[22] R. Chen, G. Xie, J. Liu, J. Wang, Z. Luo, J. Wang, and F. Zheng, "Easynet: An easy network for 3d industrial anomaly detection," in *Proceedings of the 31st ACM International Conference on Multimedia*, pp. 7038–7046, 2023.

[23] C. Bi, Y. Li, and H. Luo, "Dual-branch reconstruction network for industrial anomaly detection with rgb-d data," in *International Conference on Image, Signal Processing, and Pattern Recognition (ISPP 2024)*, vol. 13180, pp. 767–774, SPIE, 2024.

[24] Z. Li, Y. Ge, X. Wang, and L. Meng, "3d industrial anomaly detection via dual reconstruction network," *Applied Intelligence*, vol. 54, no. 20, pp. 9956–9970, 2024.

[25] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 7132–7141, 2018.

[26] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE transactions on image processing*, vol. 13, no. 4, pp. 600–612, 2004.

[27] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017.

[28] V. Zavrtanik, M. Kristan, and D. Skočaj, "Cheating depth: Enhancing 3d surface anomaly detection via depth simulation," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 2164–2172, 2024.

[29] V. Zavrtanik, M. Kristan, and D. Skočaj, "Draem-a discriminatively trained reconstruction embedding for surface anomaly detection," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 8330–8339, 2021.

[30] Y. Cheng and J. Du, "3d-pnas: 3d industrial surface anomaly synthesis with perlin noise," *arXiv preprint arXiv:2504.12856*, 2025.

[31] K. Li, B. Dai, J. Fu, and X. Hou, "Das3d: dual-modality anomaly synthesis for 3d anomaly detection," in *European Conference on Computer Vision*, pp. 148–165, Springer, 2024.

[32] V. Zavrtanik, M. Kristan, and D. Skočaj, "Keep drÆming: Discriminative 3d anomaly detection through anomaly simulation," *Pattern Recognition Letters*, vol. 181, pp. 113–119, 2024.

[33] K. Perlin, "An image synthesizer," *ACM Siggraph Computer Graphics*, vol. 19, no. 3, pp. 287–296, 1985.

[34] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.

[35] K. Long, G. Xie, L. Ma, J. Liu, and Z. Lu, "Revisiting multimodal fusion for 3d anomaly detection from an architectural perspective," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, pp. 12273–12281, 2025.

[36] H. Liu, K. Simonyan, and Y. Yang, "Darts: Differentiable architecture search," *arXiv preprint arXiv:1806.09055*, 2018.

[37] Y.-T. Liu, N. R. Pal, A. R. Marathe, and C.-T. Lin, "Weighted fuzzy dempster–shafer framework for multimodal information integration," *IEEE Transactions on Fuzzy Systems*, vol. 26, no. 1, pp. 338–352, 2017.

[38] Y. Zhao, "Anomalyhybrid: A domain-agnostic generative framework for general anomaly detection," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 3127–3136, 2025.

[39] A. Xiang, Z. Huang, X. Gao, K. Ye, and C. zhong Xu, "Bridgenet: A unified multimodal framework for bridging 2d and 3d industrial anomaly detection," 2025.

[40] M. Li, J. He, J. Li, D. Han, J. Wan, and G. Li, "Hgcf: Hierarchical geometry-color fusion for multimodal industrial anomaly detection," in *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 1023–1031, 2025.

[41] M. Cimpoi, S. Maji, I. Kokkinos, S. Mohamed, and A. Vedaldi, "Describing textures in the wild," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3606–3613, 2014.

[42] K. Long, L. Ma, J. Liu, L. Liu, and G. Xie, "Towards an incremental unified multimodal anomaly detection: Augmenting multimodal denoising from an information bottleneck perspective," *arXiv preprint arXiv:2603.02629*, 2026.

[43] N. Tishby, F. C. Pereira, and W. Bialek, "The information bottleneck method," *arXiv preprint physics/0004057*, 2000.

[44] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *First conference on language modeling*, 2024.

[45] X. Liu, J. Wang, B. Leng, and S. Zhang, "Tuned reverse distillation: Enhancing multimodal industrial anomaly detection with crossmodal tuners," 2025.

[46] M. Asad, W. Azeem, H. T. Mustafa, Y. Fang, J. Yang, Y. Zuo, and W. Liu, "Inter-modality feature prediction through multimodal fusion for 3d shape defect detection," *Neural Networks*, vol. 193, p. 108057, 2026.

[47] E. Horwitz and Y. Hoshen, "Back to the feature: classical 3d features are (almost) all you need for 3d anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2968–2977, 2023.

[48] R. B. Rusu, N. Blodow, and M. Beetz, "Fast point feature histograms (fpfh) for 3d registration," in *2009 IEEE international conference on robotics and automation*, pp. 3212–3217, IEEE, 2009.

[49] Y. Wang, J. Peng, J. Zhang, R. Yi, Y. Wang, and C. Wang, "Multimodal industrial anomaly detection via hybrid fusion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8032–8041, 2023.

[50] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2019.

[51] C. Wang, H. Zhu, J. Peng, Y. Wang, R. Yi, Y. Wu, L. Ma, and J. Zhang, "M3dm-nr: Rgb-3d noisy-resistant industrial anomaly detection via multimodal denoising," 2024.

[52] Y.-M. Chu, C. Liu, T.-I. Hsieh, H.-T. Chen, and T.-L. Liu, "Shape-guided dual-memory learning for 3d anomaly detection," in *Proceedings of the 40th International Conference on Machine Learning*, pp. 6185–6194, 2023.

[53] K. Taunk, S. De, S. Verma, and A. Swetapadma, "A brief review of nearest neighbor algorithm for learning and classification," in *2019 International Conference on Intelligent Computing and Control Systems (ICCS)*, pp. 1255–1260, 2019.

[54] S. Liu, X. Luan, and Y. Li, "Multimodal industrial anomaly detection via attention-enhanced memory-guided network," *IEEE Transactions on Multimedia*, 2025.

[55] Y. Tu, B. Zhang, L. Liu, Y. Li, J. Zhang, Y. Wang, C. Wang, and C. Zhao, "Self-supervised feature adaptation for 3d industrial anomaly detection," in *Computer Vision – ECCV 2024* (A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler, and G. Varol, eds.), (Cham), pp. 75–91, Springer Nature Switzerland, 2025.

[56] B. Jiang, Z. Zhang, D. Lin, J. Tang, and B. Luo, "Semi-supervised learning with graph learning-convolutional networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11313–11320, 2019.

[57] A. Van Den Oord, O. Vinyals, *et al.*, "Neural discrete representation learning," *Advances in neural information processing systems*, vol. 30, 2017.

[58] R. Wuken and L. Geng, "Unsupervised 3d anomaly detection method using fusion of point cloud and image features," in *2025 CAA Symposium on Fault Detection, Supervision, and Safety for Technical Processes (SAFEPROCESS)*, pp. 1–6, IEEE, 2025.

[59] Y. Lin, H. Yan, X. Tong, Y. Chang, H. Wang, Z. Zhou, S. Gao, Y. Wang, and W. Zhang, "Commonality in few: Few-shot multimodal anomaly detection via hypergraph-enhanced memory," 2025.

[60] D. Zhou, J. Huang, and B. Schölkopf, "Learning with hypergraphs: Clustering, classification, and embedding," *Advances in neural information processing systems*, vol. 19, 2006.

[61] Y. Gao, Y. Feng, S. Ji, and R. Ji, "Hgnn+: General hypergraph neural networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 3, pp. 3181–3199, 2022.

[62] M. Li, J. He, G. Li, J. Li, J. Wan, and D. Han, "Multimodal industrial anomaly detection via geometric prior," *IEEE Transactions on Circuits and Systems for Video Technology*, 2025.

[63] R. Daci, V. Renò, C. Patruno, A. Cardellicchio, A. Taleb-Ahmed, M. Leo, and C. Distante, "Cross-modal mapping and dual-branch reconstruction for 2d-3d multimodal industrial anomaly detection," *arXiv preprint arXiv:2603.03939*, 2026.

[64] C. Tao, X. Cao, and J. Du, "G2sf: Geometry-guided score fusion for multimodal industrial anomaly detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 20551–20560, 2025.

[65] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[66] Y. Pang, W. Wang, F. E. Tay, W. Liu, Y. Tian, and L. Yuan, "Masked autoencoders for point cloud self-supervised learning," in *Computer*

*Vision–ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part II*, pp. 604–621, Springer, 2022.

[67] A. Costanzino, P. Z. Ramirez, G. Lisanti, and L. Di Stefano, "Multimodal industrial anomaly detection by crossmodal feature mapping," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 17234–17243, 2024.

[68] J. Wang, Y. Niu, and B. Huang, "Fusion-restoration model for industrial multimodal anomaly detection," *Neurocomputing*, vol. 637, p. 130073, 2025.

[69] H. Cheng, J. Luo, and X. Zhang, "Multimodal industrial anomaly detection via uni-modal and cross-modal fusion," *IEEE Transactions on Industrial Informatics*, vol. 21, no. 6, pp. 5000–5010, 2025.

[70] M. Asad, W. Azeem, A. A. Malik, H. Jiang, A. Ali, J. Yang, and W. Liu, "3d-mmfn: Multi-level multimodal fusion network for 3d industrial image anomaly detection," *Advanced Engineering Informatics*, vol. 65, p. 103284, 2025.

[71] J. Liu, S. Mou, N. Gaw, and Y. Wang, "Uni-3dad: Gan-inversion aided universal 3d anomaly detection on model-free products," *Expert Systems with Applications*, vol. 272, p. 126665, 2025.

[72] M. Krichen, "Generative adversarial networks," in *2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–7, 2023.

[73] Y. Li, X. Zhou, S. Lan, W. Wang, X. Lai, and Y. Qiao, "Madflow: Multimodal difference compensation flow for multimodal anomaly detection," *Neurocomputing*, vol. 654, p. 131243, 2025.

[74] U. Ali, A. Zia, A. Rehman, U. Ramzan, Z. Hassan, T. Sattar, J. Wang, and W. Xiang, "2d_3d feature fusion via cross-modal latent synthesis and attention guided restoration for industrial anomaly detection," 2025.

[75] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "Cbam: Convolutional block attention module," in *Computer Vision – ECCV 2018* (V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, eds.), (Cham), pp. 3–19, Springer International Publishing, 2018.

[76] Y. Li, F. Liu, J. Liao, S. Tian, C.-S. Foo, and X. Yang, "Find: Few-shot anomaly inspection with normal-only multi-modal data," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 23290–23299, October 2025.

[77] W. Sui, D. Lichau, J. Lefèvre, and H. Phelippeau, "Incomplete multimodal industrial anomaly detection via cross-modal distillation," *Information Fusion*, p. 103572, 2025.

[78] W. Li, B. Zheng, X. Xu, J. Gan, F. Lu, X. Li, N. Ni, Z. Tian, X. Huang, S. Gao, *et al.*, "Multi-sensor object anomaly detection: Unifying appearance, geometry, and internal properties," *arXiv preprint arXiv:2412.14592*, 2024.

[79] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, "Towards total recall in industrial anomaly detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 14318–14328, 2022.

[80] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, "Dino: Detr with improved denoising anchor boxes for end-to-end object detection," 2022.

[81] P. Bergmann and D. Sattlegger, "Anomaly detection in 3d point clouds using deep geometric descriptors," in *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 2612–2622, 2023.

[82] A. M. Ikotun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, "K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data," *Information Sciences*, vol. 622, pp. 178–210, 2023.

[83] Q. Hu, B. Yang, L. Xie, S. Rosa, Y. Guo, Z. Wang, N. Trigoni, and A. Markham, "Randla-net: Efficient semantic segmentation of large-scale point clouds," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2020.

[84] Y. Jing, J.-X. Zhong, B. Sheil, and S. Acikgoz, "Anomaly detection of cracks in synthetic masonry arch bridge point clouds using fast point feature histograms and patchcore," *Automation in Construction*, vol. 168, p. 105766, 2024.

[85] Y. Cao, Q. Wan, W. Shen, and L. Gao, "Informative knowledge distillation for image anomaly segmentation," *Knowledge-Based Systems*, vol. 248, p. 108846, 2022.

[86] Q. Wan, L. Gao, X. Li, and L. Wen, "Industrial image anomaly localization based on gaussian clustering of pretrained feature," *IEEE Transactions on Industrial Electronics*, vol. 69, no. 6, pp. 6182–6192, 2021.

[87] Y. Cao, X. Xu, and W. Shen, "Complementary pseudo multimodal feature for point cloud anomaly detection," *Pattern Recognition*, vol. 156, p. 110761, 2024.

[88] H. Su, S. Maji, E. Kalogerakis, and E. Learned-Miller, "Multi-view convolutional neural networks for 3d shape recognition," in *Proceedings of the IEEE international conference on computer vision*, pp. 945–953, 2015.

[89] H. Liang, J. Zhou, X. Chen, T. Dai, J. Wang, and C. Gao, "Fence theorem: Towards dual-objective semantic-structure isolation in preprocessing phase for 3d anomaly detection," 2025.

[90] Y. Cheng, Y. Sun, H. Zhang, W. Shen, and Y. Cao, "Towards high-resolution 3d anomaly detection: A scalable dataset and real-time framework for subtle industrial defects," 2025.

[91] C. Zhou, J. Wan, and X. Zhao, "Real-time industrial anomaly detection via sparse reconstruction," in *2024 IEEE International Conference on Real-time Computing and Robotics (RCAR)*, pp. 552–557, 2024.

[92] Y. Jing, W. Lin, B. Sheil, and S. Acikgoz, "3d multimodal feature for infrastructure anomaly detection," *Automation in Construction*, vol. 178, p. 106388, 2025.

[93] J. Du and D. Chen, "Position: Untrained machine learning for anomaly detection by using 3d point cloud data," 2025.

[94] C. Tao and J. Du, "Pointsgrade: Sparse learning with graph representation for anomaly detection by using unstructured 3d point cloud data," *IISE Transactions*, vol. 57, no. 2, pp. 131–144, 2025.

[95] J. Qin, C. Gu, J. Yu, and C. Zhang, "Teacher–student network for 3d point cloud anomaly detection with few normal samples," *Expert Syst. Appl.*, vol. 228, Oct. 2023.

[96] J. Liu, G. Xie, R. Chen, X. Li, J. Wang, Y. Liu, C. Wang, and F. Zheng, "Real3d-ad: A dataset of point cloud anomaly detection," *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, vol. 36, pp. 30402–30415, 2023.

[97] R. C. Bolles and M. A. Fischler, "A ransac-based approach to model fitting and its application to finding cylinders in range data," in *Proceedings of the 7th International Joint Conference on Artificial Intelligence - Volume 2*, IJCAI'81, (San Francisco, CA, USA), p. 637–643, Morgan Kaufmann Publishers Inc., 1981.

[98] B. Zhao, Q. Xiong, X. Zhang, J. Guo, Q. Liu, X. Xing, and X. Xu, "Pointcore: Efficient unsupervised point cloud anomaly detector using local-global features," *arXiv preprint arXiv:2403.01804*, 2024.

[99] H. Zhu, G. Xie, C. Hou, T. Dai, C. Gao, J. Wang, and L. Shen, "Towards high-resolution 3d anomaly detection via group-level feature contrastive learning," in *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 4680–4689, 2024.

[100] H. Liang, G. Xie, C. Hou, B. Wang, C. Gao, and J. Wang, "Look inside for more: Internal spatial modality perception for 3d anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, pp. 5146–5154, 2025.

[101] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*, pp. 6105–6114, PMLR, 2019.

[102] Y. Cheng, Y. Cao, D. Wang, W. Shen, and W. Li, "Boosting global-local feature matching via anomaly synthesis for multi-class point cloud anomaly detection," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 12560–12571, 2025.

[103] Y. Yu, Z. Chen, X. Xu, L. Zhang, H. Yang, Y. Nie, and S. He, "Registration is a powerful rotation-invariance learner for 3d anomaly detection," *arXiv preprint arXiv:2510.16865*, 2025.

[104] Z. Zhang, B.-S. Hua, and S.-K. Yeung, "Riconv++: Effective rotation invariant convolutions for 3d point clouds deep learning," *International Journal of Computer Vision*, vol. 130, no. 5, pp. 1228–1243, 2022.

[105] H. Thomas, C. R. Qi, J.-E. Deschaud, B. Marcotegui, F. Goulette, and L. J. Guibas, "Kpconv: Flexible and deformable convolution for point clouds," *Proceedings of the IEEE International Conference on Computer Vision*, 2019.

[106] Z. Qin, H. Yu, C. Wang, Y. Guo, Y. Peng, and K. Xu, "Geometric transformer for fast and robust point cloud registration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11143–11152, June 2022.

[107] H. Liang, J. Zhou, C. Gao, B. Guo, J. Wang, and L. Shen, "A lightweight 3d anomaly detection method with rotationally invariant features," 2025.

[108] Y. Liu, C. Zhang, and Y. Yang, "Template3d-ad: Point cloud template matching method based on center points for 3d anomaly detection," in *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pp. 1630–1638, 2025.

[109] J. Wang, X. Wang, R. Hao, H. Yin, B. Huang, X. Xu, and J. Liu, "Incremental template neighborhood matching for 3d anomaly detection," *Neurocomputing*, vol. 581, p. 127483, 2024.

[110] X. Cao, C. Tao, and J. Du, "Deep subspace learning for surface anomaly classification based on 3d point cloud data," 2025.

[111] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "Pointnet: Deep learning on point sets for 3d classification and segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 652–660, 2017.

[112] J. Liao, X. Xu, Y. Su, R.-C. Tu, Y. Liu, D. Tao, and X. Yang, "Robust distribution alignment for industrial anomaly detection under distribution shift," 2025.

[113] Y. Su, X. Xu, T. Li, and K. Jia, "Revisiting realistic test-time training: Sequential inference and adaptation by anchored clustering regularized self-training," 2023.

[114] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Advances in Neural Information Processing Systems* (C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Weinberger, eds.), vol. 26, Curran Associates, Inc., 2013.

[115] A. Bhunia, C. Li, and H. Bilen, "Looking 3d: Anomaly detection with 2d-3d alignment," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 17263–17272, 2024.

[116] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2117–2125, 2017.

[117] Y. Sun, Y. Cheng, Y. Cao, Y. Zhang, and W. Shen, "Multi-view reconstruction with global context for 3d anomaly detection," 2025.

[118] M. Asad, W. Azeem, H. Jiang, H. T. Mustafa, J. Yang, and W. Liu, "2m3df: Advancing 3d industrial defect detection with multi-perspective multimodal fusion network," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 7, pp. 6803–6815, 2025.

[119] Y. Yang, Z. Fu, Y. Li, L. Wang, S. Yu, and J. Xiao, "Probing 3d anomalies via multi-view registration and dual-residual analysis," *Available at SSRN 5868807*.

[120] Z. Wang, K. Hotta, K. Kamide, Y. Zou, J. Qin, C. Zhang, and J. Yu, "Dmp-3dad: Cross-category 3d anomaly detection via realistic depth map projection with few normal samples," *arXiv preprint arXiv:2602.10806*, 2026.

[121] Y. Yang, C. Feng, Y. Shen, and D. Tian, "Foldingnet: Point cloud auto-encoder via deep grid deformation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 206–215, 2018.

[122] M. Masuda, R. Hachiuma, R. Fujii, H. Saito, and Y. Sekikawa, "Toward unsupervised 3d point cloud anomaly detection using variational autoencoder," in *2021 IEEE International Conference on Image Processing (ICIP)*, pp. 3118–3122, IEEE, 2021.

[123] W. Li, X. Xu, Y. Gu, B. Zheng, S. Gao, and Y. Wu, "Towards scalable 3d anomaly detection and localization: A benchmark via 3d anomaly synthesis and a self-supervised learning network," 2023.

[124] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," 2021.

[125] H. Liang, J. Zhang, T. Dai, L. Shen, J. Wang, and C. Gao, "Taming anomalies with down-up sampling networks: Group center preserving reconstruction for 3d anomaly detection," in *Proceedings of the 33rd ACM International Conference on Multimedia (MM2025)*, 2025.

[126] H. Li, Y. Niu, H. Yin, Y. Mo, Y. Liu, B. Huang, R. Wu, and J. Liu, "Daup: Enhancing point cloud homogeneity for 3d industrial anomaly detection via density-aware point cloud upsampling," *Advanced Engineering Informatics*, vol. 62, p. 102823, 2024.

[127] X. Cao, C. Tao, Y. Cheng, and J. Du, "Iaenet: An importance-aware ensemble model for 3d point cloud-based anomaly detection," *Information Fusion*, p. 104097, 2025.

[128] X. Yu, L. Tang, Y. Rao, T. Huang, J. Zhou, and J. Lu, "Point-bert: Pre-training 3d point cloud transformers with masked point modeling," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[129] Z. You, L. Cui, Y. Shen, K. Yang, X. Lu, Y. Zheng, and X. Le, "A unified model for multi-class anomaly detection," *Advances in Neural Information Processing Systems*, vol. 35, pp. 4571–4584, 2022.

[130] J. Cheng, C. Gao, J. Zhou, J. Wen, T. Dai, and J. Wang, "Mc3d-ad: A unified geometry-aware reconstruction model for multi-category 3d anomaly detection," in *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25* (J. Kwok, ed.), pp. 837–845, International Joint Conferences on Artificial Intelligence Organization, 8 2025. Main Track.

[131] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[132] H. Lu, H. Liang, J. Zhang, C. Hu, J. Wang, and C. Gao, "C3d-ad: Toward continual 3d anomaly detection via kernel attention with learnable advisor," 2025.

[133] L. Wu, J. Ning, and Q. Zou, "Anomaly-aware siamese comparative transformer for 3d anomaly detection," *SSRN Electronic Journal*, October 2025.

[134] D.-C. Hoang, P. X. Tan, A.-N. Nguyen, T. H. A. Duong, T.-M. Huynh, M.-A. Nguyen, D.-M. Nguyen, M.-D. Cao, D.-H. Ngo, M.-Q. Vu, *et al.*, "Vote3d-ad: Unsupervised point cloud anomaly localization via varied defect synthesis and differentiable vote-clustering," *Alexandria Engineering Journal*, vol. 137, pp. 171–193, 2026.

[135] Z. Zhou, L. Wang, N. Fang, Z. Wang, L. Qiu, and S. Zhang, "R3d-ad: Reconstruction via diffusion for 3d anomaly detection," in *European conference on computer vision*, pp. 91–107, Springer, 2024.

[136] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.

[137] J. Ye, W. Zhao, X. Yang, G. Cheng, and K. Huang, "Po3ad: Predicting point offsets toward better 3d point cloud anomaly detection," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 1353–1362, 2025.

[138] C. Choy, J. Gwak, and S. Savarese, "4d spatio-temporal convnets: Minkowski convolutional neural networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 3075–3084, 2019.

[139] C. Choy, J. Park, and V. Koltun, "Fully convolutional geometric features," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 8958–8966, 2019.

[140] H. Liang, A. Wang, J. Zhou, X. Jin, C. Gao, and J. Wang, "Examining the source of defects from a mechanical perspective for 3d anomaly detection," in *arXiv*, 2025.

[141] Y. Zha, X. Yuerong, C. Fan, Y. Wang, T. Dai, K. Chen, and S.-T. Xia, "Casl: Curvature-augmented self-supervised learning for 3d anomaly detection," *arXiv preprint arXiv:2511.12909*, 2025.

[142] Y. Huang and Z. Qi, "Curvature-fused feature extractor for accelerated point cloud anomaly detection," *IEEE Access*, 2026.

[143] L. Liu, H. Liang, C. Gao, W. Zhuo, L. Shen, and J. Wang, "Uca-3dal: A unified contrastive framework with test-time adaptation for robust 3d anomaly localization," in *Arxiv*, 2025.

[144] B. Ma, Z. Han, Y.-S. Liu, and M. Zwicker, "Neural-pull: Learning signed distance functions from point clouds by learning to pull space onto surfaces," *arXiv preprint arXiv:2011.13495*, 2020.

[145] B. Ma, Y.-S. Liu, M. Zwicker, and Z. Han, "Surface reconstruction from point clouds by learning predictive context priors," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6326–6337, 2022.

[146] X. Cao, C. Tao, and J. Du, "3d-csad: Untrained 3d anomaly detection for complex manufacturing surfaces," *arXiv preprint arXiv:2404.07748*, 2024.

[147] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?," *Journal of the ACM (JACM)*, vol. 58, no. 3, pp. 1–37, 2011.

[148] B. Zheng, J. Gan, X. Xu, W. Li, X. Huang, N. Ni, and Y. Wu, "Bridging 3d anomaly localization and repair via high-quality continuous geometric representation," 2025.

[149] J. Wang, H. Xu, X. Chen, H. Xu, Y. Huang, X. Ding, and X. Tu, "Exploiting point-language models with dual-prompts for 3d anomaly detection," 2025.

[150] Z. Wang, K. Hotta, K. Kamide, Y. Zou, C. Zhang, and J. Yu, "3dkeyad: High-resolution 3d point cloud anomaly detection via keypoint-guided point clustering," *arXiv preprint arXiv:2507.13110*, 2025.

[151] G. Xu, J. Du, and H. Yang, "3d-sonar: Self-organizing network for 3d anomaly ranking," *arXiv preprint arXiv:2601.09294*, 2026.

[152] L. Li, "Unsupervised feature metric-based multimodal anomaly detection method," in *2025 5th International Conference on Artificial Intelligence, Big Data and Algorithms (CAIBDA)*, pp. 1597–1602, IEEE, 2025.

[153] Z. Wang, K. Hotta, Y. Zou, Y. Ding, C. Zhang, and J. Yu, "Boosting high-resolution 3d point cloud anomaly detection with geometric constraints," in *Eighteenth International Conference on Machine Vision (ICMV 2025)*, vol. 14114, pp. 473–480, SPIE, 2026.

[154] OpenAI, J. Achiam, . Steven Adler, W. Zhuk, and B. Zoph, "Gpt-4 technical report," 2024.

[155] Y. Cao, X. Xu, C. Sun, X. Huang, and W. Shen, "Towards generic anomaly detection and understanding: Large-scale visual-linguistic model (gpt-4v) takes the lead," 2023.

[156] Q. Zhou, J. Yan, S. He, W. Meng, and J. Chen, "Pointad: Comprehending 3d anomalies from points and pixels for zero-shot 3d anomaly detection," in *Advances in Neural Information Processing Systems* (A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak,

and C. Zhang, eds.), vol. 37, pp. 84866–84896, Curran Associates, Inc., 2024.
[157] R. Zhang, Z. Guo, W. Zhang, K. Li, X. Miao, B. Cui, Y. Qiao, P. Gao, and H. Li, "Pointclip: Point cloud understanding by clip," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8552–8562, 2022.
[158] Q. Zhou, S. He, J. Yan, W. Meng, and J. Chen, "Pointad+: Learning hierarchical representations for zero-shot 3d anomaly detection," *arXiv preprint arXiv:2509.03277*, 2025.
[159] Y. Wang, K.-C. Peng, and Y. Fu, "Towards zero-shot 3d anomaly localization," in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 1447–1456, IEEE, 2025.
[160] Z. Deng, A. Liu, and Y. Wang, "Gs-clip: Zero-shot 3d anomaly detection by geometry-aware prompt and synergistic view representation learning," *arXiv preprint arXiv:2602.19206*, 2026.
[161] Y. Cheng, Y. Cao, G. Xie, Z. Lu, and W. Shen, "Towards zero-shot point cloud anomaly detection: A multi-view projection framework," *arXiv preprint arXiv:2409.13162*, 2024.
[162] Z. Zuo, J. Dong, Y. Wu, Y. Qu, and Z. Wu, "Clip3d-ad: Extending clip for 3d few-shot anomaly detection with multi-view images generation," 2024.
[163] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*, pp. 8748–8763, PmLR, 2021.
[164] X. Xu, Y. Cao, H. Zhang, N. Sang, and X. Huang, "Customizing visual-language foundation models for multi-modal anomaly detection and reasoning," *arXiv preprint arXiv:2403.11083*, 2024.
[165] G. Li, T. Chen, M. Zhou, M. Li, D. Han, and J. Wan, "Mcl-ad: Multimodal collaboration learning for zero-shot 3d anomaly detection," *arXiv preprint arXiv:2509.10282*, 2025.
[166] X. Li, F. Xue, and Y. Zhou, "Musc-v2: Zero-shot multimodal industrial anomaly classification and segmentation with mutual scoring of unlabeled samples," *arXiv preprint arXiv:2511.10047*, 2025.
[167] Q. Zhou, W. Li, L. Jiang, G. Wang, G. Zhou, S. Zhang, and H. Zhao, "Pad: A dataset and benchmark for pose-agnostic anomaly detection," in *Advances in Neural Information Processing Systems* (A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, eds.), vol. 36, pp. 44558–44571, Curran Associates, Inc., 2023.
[168] B. Cao, A. Araujo, and J. Sim, "Unifying deep local and global features for image search," in *Computer Vision – ECCV 2020* (A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, eds.), (Cham), pp. 726–743, Springer International Publishing, 2020.
[169] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "Nerf: representing scenes as neural radiance fields for view synthesis," *Commun. ACM*, vol. 65, p. 99–106, Dec. 2021.
[170] L. Yen-Chen, P. Florence, J. T. Barron, A. Rodriguez, P. Isola, and T.-Y. Lin, "inerf: Inverting neural radiance fields for pose estimation," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 1323–1330, 2021.
[171] M. Kruse, M. Rudolph, D. Woiwode, and B. Rosenhahn, "Splatpose & detect: Pose-agnostic 3d anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 3950–3960, June 2024.
[172] Y. Liu, Y. S. Hu, Y. Chen, and J. Zelek, "Splatpose+: Real-time image-based pose-agnostic 3d anomaly detection," in *Computer Vision – ECCV 2024 Workshops* (A. Del Bue, C. Canton, J. Pont-Tuset, and T. Tommasi, eds.), (Cham), pp. 378–391, Springer Nature Switzerland, 2025.
[173] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3d gaussian splatting for real-time radiance field rendering," 2023.
[174] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4104–4113, 2016.
[175] K. Yang, J. Cao, Z. Bai, Z. Su, and A. Tagliasacchi, "Piad: Pose and illumination agnostic anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4734–4743, June 2025.
[176] Y. Wang, R. Yi, X. Wang, J. Hao, Y. Wang, K. Fan, Y. Chen, Z. Zhang, G. Jiang, W. Zhu, *et al.*, "Pose-agnostic anomaly detection with retinex-based illumination alignment," *Available at SSRN 6055450*.
[177] W. Wu, J. Weng, P. Zhang, X. Wang, W. Yang, and J. Jiang, "Uretinex-net: Retinex-based deep unfolding network for low-light image enhancement," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5891–5900, 2022.
[178] P. Bergmann, X. Jin, D. Sattlegger, and C. Steger, "The mvtec 3d-ad dataset for unsupervised 3d anomaly detection and localization," in *Proceedings of the 17th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2022) - Volume 5: VISAPP*, pp. 202–213, INSTICC, SciTePress, 2022.
[179] L. Bonfiglioli, M. Toschi, D. Silvestri, N. Fioraio, and D. De Gregorio, "The eyecandies dataset for unsupervised multimodal anomaly detection and localization," in *Proceedings of the 16th Asian Conference on Computer Vision (ACCV2022)*, 2022. ACCV.
[180] J. Qin, C. Gu, J. Yu, and C. Zhang, "Image-pointcloud fusion based anomaly detection using pd-real dataset," 2023.
[181] C. Wang, W. Zhu, B.-B. Gao, Z. Gan, J. Zhang, Z. Gu, S. Qian, M. Chen, and L. Ma, "Real-iad: A real-world multi-view dataset for benchmarking versatile industrial anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22883–22892, 2024.
[182] P. McHard, F. P. Audonnet, O. Summerell, S. Andraos, P. Henderson, and G. Aragon-Camarasa, "3d-adam: A dataset for 3d anomaly detection in additive manufacturing," 2025.
[183] A. Costanzino, P. Z. Ramirez, L. Lella, M. Ragaglia, A. Oliva, G. Lisanti, and L. D. Stefano, "Sim3d: Single-instance multiview multimodal and multisetup 3d anomaly detection benchmark," 2025.
[184] Y. Cheng, Y. Sun, H. Zhang, W. Shen, and Y. Cao, "Towards high-resolution 3d anomaly detection: A scalable dataset and real-time framework for subtle industrial defects," 2025.
[185] W. Zhu, L. Wang, Z. Zhou, C. Wang, Y. Pan, R. Zhang, Z. Chen, L. Cheng, B.-B. Gao, J. Zhang, Z. Gan, Y. Wang, Y. Chen, S. Qian, M. Chi, B. Peng, and L. Ma, "Real-iad d3: A real-world 2d/pseudo-3d/3d dataset for industrial anomaly detection," 2025.
[186] W. Zhu, C. Wang, B.-B. Gao, J. Zhang, G. Jiang, J. Hu, Z. Gan, L. Wang, Z. Zhou, L. Cheng, Y. Pan, B. Peng, M. Chi, and L. Ma, "Real-iad variety: Pushing industrial anomaly detection dataset to a modern era," 2025.
[187] B. Guo, Q. Zuo, and R. Yu, "Point cloud segmentation of integrated circuits package substrates surface defects using causal inference: Dataset construction and methodology," 2025.
[188] B. Guo, H. Li, R. Yu, H. Liang, and J. Wang, "Iec3d-ad: A 3d dataset of industrial equipment components for unsupervised point cloud anomaly detection," 2025.
[189] X. Jiang, J. Li, H. Deng, Y. Liu, B.-B. Gao, Y. Zhou, J. Li, C. Wang, and F. Zheng, "Mmad: The comprehensive benchmark for multimodal large language models in industrial anomaly detection," in *The Thirteenth International Conference on Learning Representations*, 2025.
[190] X. Jiang, G. Xie, J. Wang, Y. Liu, C. Wang, F. Zheng, and Y. Jin, "A survey of visual sensory anomaly detection," *arXiv preprint arXiv:2202.07006*, 2022.

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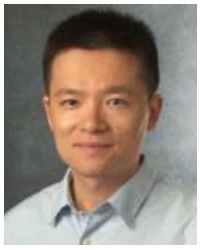

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